RESEARCH ARTICLE

Quantification of Ghana's historical mining expansion: Back-casting from a recent mine inventory

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ARTICLE HISTORY
Compiled November 6, 2026

ABSTRACT
Ghana has witnessed a rapid expansion of mining, especially in its south-western forest zone. This expansion is visible and measurable from space using remote sensing, and introduces profound changes for people living in these areas. Accurately quantifying the timing and extent of mining is important to evaluate policies, sustainability, and development outcomes but is complicated by cloud cover and poor-quality imagery in earlier Landsat years (pre-2013). In this study, I am developing annual mining footprints for all of Ghana based on high-resolution mapping of mines in 2019. Using Landsat imagery from 2000 through 2019 (27 spectral, index, texture and terrain bands), training a national random forest (RF) classifier to detect mining, vegetation, and water, and applying backward transition rules to prevent implausible class reversions, I estimate historical mining expansion within known mining boundaries. Under spatially blocked cross-validation with a variogram-derived buffer, the model reaches an overall accuracy of 0.87 and a mining F_1 of 0.78, with zone-specific models not performing better. I validate my approach by drawing 1000 Monte Carlo simulations from a bootstrapped ensemble, and comparing industrial mining onset to industry statistics and an independent convolutional neural network (CNN). I find that constraining classification to known mining areas enables an improved assessment of mining-related land conversion which includes mine infrastructure and tailings, while backward transition rules prevented implausible pixel transitions between landcover classes due to low data quality. Results indicate Ghana's mining area expanded from approximately 1 623 ha in 2000 to 53 391 ha in 2019, with peak expansion occurring during 2012 and 2013, and from 2016 onwards. My approach demonstrates that recent improvements in remote sensing can be used to train classification models for better historical analyses of lower resolution and lower quality imagery. The results show how recent high-resolution imagery can contribute to improving inferences from existing low-resolution imagery, enabling time-series analysis before high-resolution imagery was available.

KEYWORDS
Mining expansion; Landsat time series; random forest; spatial cross-validation; artisanal and small-scale mining; Ghana

1. Introduction

The accurate assessment of mining related landcover changes on the earth's surface is of paramount importance to understanding and explaining related ecological, societal, and economic consequences, as well as policy, sustainability, and development

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outcomes. Mining activities are concentrated around mineral deposits and promise economic gains and long-term development (Andrews, 2018; Knutsen, Kotsadam, Olsen, & Wig, 2017) but might also cause adverse ecological and societal effects if not managed well (A. K. Mensah et al., 2015). For mineral rich countries such as Ghana, where mining makes up around 17% of government revenue and a majority of exports (Siachiwena, 2025, p. 3), correctly monitoring the expansion and impacts of mining for its populations, environment, and economy is crucial. However, no annual mining footpint data combining both major large-scale operations and smaller ventures, is currently available. This study's objective is therefore to map Ghana's historical mining expansion and create a time series of extent polygons for further analysis. My annual mining data is crucial for assessing the combined effects of mining, as well as differentiating between small-scale and industrial effects in future mining impact studies. My work first introduces challenges in mining landcover classification before suggesting relevant methodologies to address some of these shortcomings at fine grained temporal and disaggregated level. I suggest a historical analysis of landcover change using supervised learning and focusing classification on known mining areas.

Mining sites in Ghana provide a relevant and instructive example as no spatial and temporal data combining industrial and "small-scale" operations for the last decades exist. Spanning approximately 4.5°N to 11.5°N latitude (Fig. 1), the country extents over multiple agro-ecological zones with distinct climatic, vegetative, and land use characteristics that challenge accurate remote sensing-based studies (Yamba et al., 2023). The country experienced rapid expansion of mining activities following liberalization policies in 1986, with artisanal and small-scale gold mining intensifying dramatically in response to rising commodity prices (Andrews, 2018; McQuilken & Hilson, 2016; Wolford et al., 2024) and large mining operations buying concessions. Mining is concentrated in Ghana's southwestern forest zone where Birimian and Tarkwaian gold deposits have been found (Milési et al., 1991), and where historically, forest and agriculture were the prominent means of subsistence (Wills, 1962 cited in Barenblitt et al., 2021, p.3). Stark contrasts between open-pits and dense forests invite for the use of remote sensing methods (Barenblitt et al., 2021) but persistent cloud cover in the region demands for multi-year seasonal compositing to achieve adequate image quality (Flood, 2013; Phan, Kuch, & Lehnert, 2020).

Mining sites also exhibit spectral signatures that overlap substantially with urban areas, bare agricultural soil, natural rock, shallow river beds, and construction sites (Barenblitt et al., 2021; Gallwey, Robiati, Coggan, Vogt, & Eyre, 2020; Snapir, Simms, & Waine, 2017). Furthermore, mining landscapes include tailings ponds, processing infrastructure, access roads, and worker settlements, which all represent conversion from prior land cover yet may be classified as non-mining by standard algorithms. Previous studies addressing mining expansion in sub-Saharan Africa have either focused on industrial operations verified through concession records (Ahmed, Bryant, & Edwards, 2021), employed single random forest models across ecologically diverse study areas (Barenblitt et al., 2021), or applied deep learning approaches to parts of a country in years where high resolution imagery ($<$10m resolution) is available (Gallwey et al., 2020; Nursamsi, Phinn, Levin, Luskin, & Sonter, 2024). These studies point to inherent challenges regarding class confusion, particularly between mining, shallow water bodies, and urban land cover (Barenblitt et al., 2021; Zhang et al., 2026) and data quality more generally (Kattenborn, Leitloff, Schiefer, & Hinz, 2021; Nursamsi et al., 2024). There is also a noticeable rift between industrial and artisanal mining

in the literature, which are often studied seperately[1] although they both alter landscapes and share many sympiotic relationships (Arthur-Holmes & Abrefa Busia, 2022; Brunnschweiler, Karapetyan, & Lujala, 2024; Ros-Tonen et al., 2021). Several existing remote sensing studies have further focused on the comparison of few images (Obodai, Adjei, Odai, & Lumor, 2019), or studied limited periods of time (Forkuor, Ullmann, & Griesbeck, 2020; Gallwey et al., 2020), producing conflicting results regarding the timing and extent of mining activities in Ghana.

My study leverages the mining polygons of Maus et al. (2022) to develop an improved framework for quantifying mining expansion in Ghana from 2000 to 2019. I propose combining two methodological approaches: 1) constraining the classification domain to known mining areas, and 2) using one-way transition rules (henceforth "transition constraints") to increase consistency and accuracy. The resulting national, and district-level time series of mining area provides spatially and temporally disaggregated estimates suitable for integration with socioeconomic analyses of mining impacts. The RF model is validated conservatively using spatial cross-validation (CV), area of applicability (AoA) mapping, and against a CNN comparing mining onset at industrial concessions. My results are relevant for replication in other contexts.

[1] See e.g. Barenblitt et al. (2021), Gallwey et al. (2020), Ahmed et al. (2021) or Ros-Tonen, Aggrey, Somuah, and Derkyi (2021).

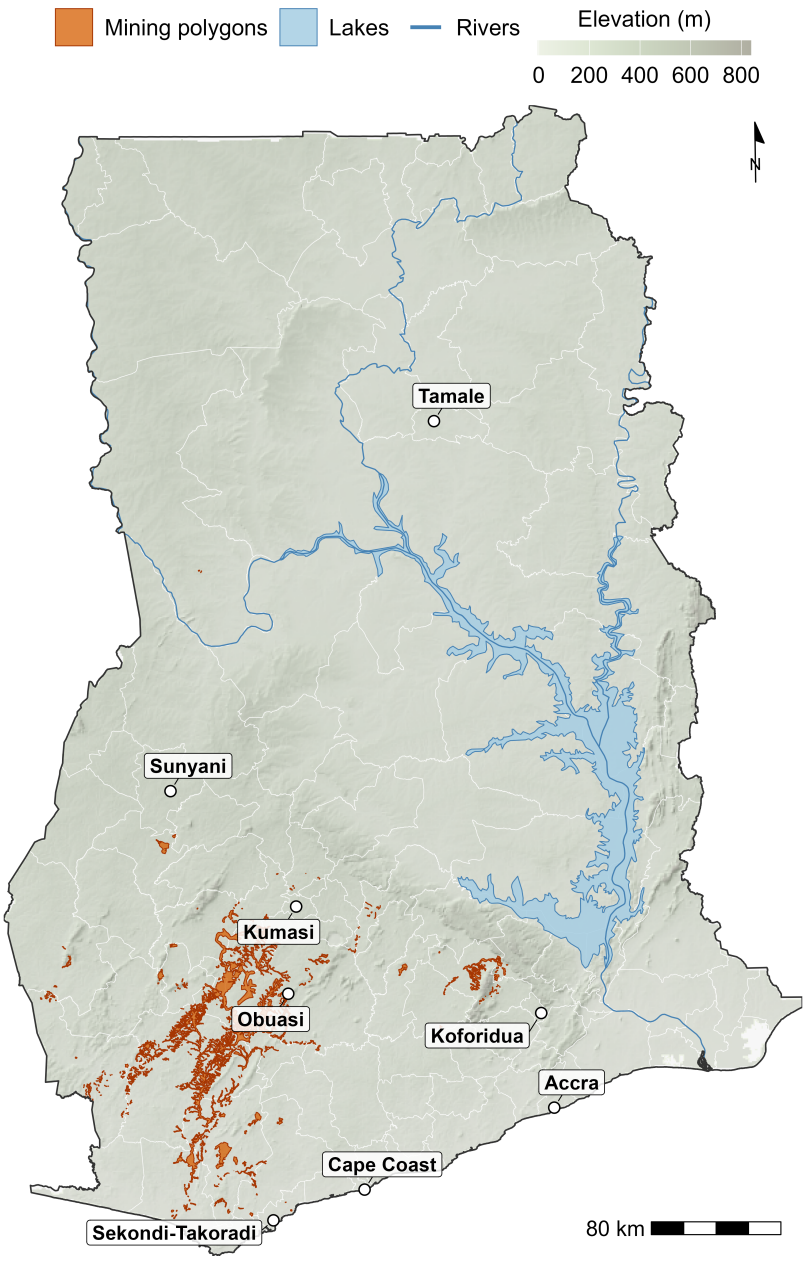


Figure 1. Study area showing mining polygons from Maus et al. (2022). Mining activity is concentrated in southwestern Ghana. Ghana's districts, elevation, and water overlayed.

2. Methods and Data

Especially challenging for the creation of consistent time series data across differing sensors (Landsat-Sentinel etc.) is the data quality, resolution, and availability of imagery, impacting choice of methods such as random forest (RF) classifiers or deep learning approaches such as a U-Net or standard Convolutional Neural Network (CNN). Ma et al. (2019, pp.170,172), Ball, Anderson, and Chan (2017), and Kattenborn et al. (2021, p.25) point out that RF perform well with few training samples, while deep learning approaches require finer resolution imagery (10 m or lower) and detailed training data

to learn mining features appropriately but handle feature engineering and variables implicitly. Nursamsi et al. (2024, pp.7-9) showcase that in areas of high spectral confusion such as polluted water bodies and forest mining sites, pixel based RF can produce high accuracy classification, although country context and sensor type are decicive for the choice of method.

The publication of global mining polygons by Maus et al. (2022, 2020) introduced manually digitized and verified mining polygons for 2019, improving on previous mining datasets derived from concession boundaries or point locations (Maus et al., 2022). The dataset also captures both industrial and artisanal operations, providing unprecedented reference data for temporal analyses. Maus et al. (2022) mining data enables both the focused sampling of datapoints and the training of classification models using the historical Landsat acrchive[2] through Google Earth Engine's cloud computing platform (Gorelick et al., 2017). Since random forest classifiers show good robustness to noise, low data quality, ability to handle spectral information of diverse landscapes, and resistance to overfitting (Belgiu & Drăguţ, 2016; Breiman, 2001; Gislason, Benediktsson, & Sveinsson, 2006; Phiri & Morgenroth, 2017; Rodriguez-Galiano, Ghimire, Rogan, Chica-Olmo, & Rigol-Sanchez, 2012), I decided to train an ensemble of RF classifiers for my historical time-series analyses.

As my study period covers coarse resolution (30m) years with low data quality, and spans multiple sensor shifts, my study opts for a RF approach over deep learning as they require rich training data and high resolution imagery which is only available in more recent years (Gallwey et al., 2020; Kattenborn et al., 2021; Nursamsi et al., 2024). I validate my mining onset results against a CNN for comparison.

2.1. *Satellite imagery and composite generation*

For my analysis, I created a time series of cloud-masked satellite composites spanning 2000 to 2019 at 30 m resolution using Google Earth Engine (Gorelick et al., 2017)[3]. Since surface reflectance signatures vary significantly between wet and dry seasons in Ghana (Flood, 2013), I restricted image aquisition to the extended dry season (November through April) to ensure spectral consistency across years.

To address persistent cloud contamination in Ghana's South-West (Barenblitt et al., 2021), I opted for medoid compositing within three-year temporal windows centered on each target year to maximize cloud-free coverage while maintaining spectral consistency (Flood, 2013; Phan et al., 2020).

2.2. *Variables and feature selection*

Each composite was created with 40 bands initially, including six categories of bands, plus quality control bands (See Table 9). Spectral bands, spectral indices, water indices, urban index, and topographic bands information was used as specified by the latest literature. After dropping quality ("housekeeping") bands, I used all 27 remaining bands for training of the random forest classifier. For validation, I conducted a comprehensive feature analysis, comparing data- and literature-driven selection for the national model (See appendix C for full data report).

[2]Longest continuous and publicly available record of medium-resolution multispectral imagery, enabling retrospective analysis of land cover change at regional and global scale (Potapov et al., 2022; Roy et al., 2016).

[3]The highest possible resolution available for free and for the entire study period.

2.3. *Training data collection*

To train the classifier on the Landsat composite, I sampled and validated 1409 training points manually using the 2019 composite as a reference layer in GEE (Gorelick et al., 2017), with NDVI, RGB, and BSI for visual interpretation. I orginally sampled from seven landcover classes (bare soil, water, dense forest, shrubland, flooded vegetation, cropland, built-up) according to ESA World Cover (Zanaga et al., 2021) data layer before aggregating to three target classes for classification: Water (including tailings ponds), Mining (bare soil, built-up), and Vegetation (cropland, dense forest, shrubland, flooded vegetation). This classification scheme is appropriate given my focus on mining expansion within known mining areas, where the primary distinction is between converted (mining-related) and unconverted (vegetation) land cover. I sampled stratified across three buffers, inside and around (1-2 km; 2-5 km) the Maus et al. (2022) mining poplygons to ensure spectral representation (Duckham, Sun, & Worboys, 2024, p. 388). Underrepresented classes (Like water around northern mining sites) were sampled across the wider ecological zones, leaving minimum one kilometer around training points to ensure spatial represenation[4].

2.4. *Random forest classification*

I trained one national ensemble random forest model for my study area using all training points and the `ranger` package in R (Wright & Ziegler, 2017)[5]. Following recommendations by Belgiu and Drăguţ (2016) and Svoboda, Štych, Laštovička, Paluba, and Kobliuk (2022), the ensemble comprised five models with 500 trees each. I followed hyperparameter recommendations by Gislason et al. (2006) using a sample fraction of 0.6 in each RF tree for bootstrap diversity. In addition to the classification raster, I also saved the class probability rasters for validation (simulations).

Class imbalance was addressed using square-root inverse-frequence class weights as Buda, Maki, and Mazurowski (2018) specify. The mining class typically represented 15 to 25% of training samples depending on zone, with vegetation dominating in forest areas and water being the minority class throughout.

For validation of accuracy, I report a full national confusion matrix per class, computing overall accuracy, as well as precision and recall curves with per class and zone F1 scores (Olofsson et al., 2014; Stehman & Foody, 2019; Tharwat, 2020).

2.5. *Transition constraints*

I trained my ensemble model on the full Ghana 2019 composite and classified landcover backward to earlier years (2000 through 2018), generating probability scenarios for each class at national and district level. Since this "unconstrained" time series from the majority ensemble vote allows for any landcover transition between years, I then apply 'temporal transition rules' on the unconstrained classification rasters so that going back in time, pixels classified as Bare/Mining or Water in 2019 can only transition to Vegetation in earlier years, never vice versa. This enforces the ecological assumption that mining area is increasing and that vegetation does not instantaneously regenerate after mining disturbance. This assumption is reasonable for my study period given that: (a) mining expansion in Ghana has been broadly increasing since 2000 (Aikins, 2024;

[4]Note that some training points are naturally clustered at less than 1 km, like mining points inside concessions.
[5]See Appendix C 2 for modelling choices and tests across ecologically diverse areas.

Asori et al., 2022; McQuilken & Hilson, 2016); (b) active transformation of mining land to vegetation is rare and slow (Barenblitt et al., 2021); and (c) the 2019 polygons capture currently active operations, not historical, abandoned mines.

I treat the transition-constrained approach as the primary method for quantifying mining expansion, as it prevents ecologically implausible transitions (e.g., dense forest appearing one year after active mining) that can arise from inter-annual classification errors, especially considering the poor image-quality in earlier Landsat years (Pre-2013). I assess the transition rules by extracting class probabilities for N=1000 simulations of Ghana's mining expansion per pixel over time and comparing confidence intervals. I display and discuss the district level and national mining area estimates for my study period. To assess whether my contraint approach is too restrictive I compare mining onset for major industrial operations from industry statistics and a CNN (See appendix E and F).

2.6. *Mining area quantification*

Classified output rasters were spatially intersected with Ghana district boundaries from the Integrated Public Use Microdata Series (Ruggles et al., 2025) to generate both national and district-level mining area statistics. Within the mining polygon boundaries of Maus et al. (2022), I report:

- Total mining-related area (Bare/Mining + Water classes) within the mining polygons in ha, both on national and top five district level
- Year-over-year change in mining area nationally and per top 5 districts

By constraining analysis to the validated mining polygons, my approach captures comprehensive mining impacts including infrastructure, tailings, and cleared areas development that previous studies may have classified as non-mining land cover (Barenblitt et al., 2021). Mining activities that occurred outside the mining polygons of Maus et al. (2022), under tree canopy, or operations that have been abandoned will not be detected by my approach. However, given the increasing gold prices and intensification of mining in Ghana, my assumption of expanding mines should hold true and contribute with relevant estimates.

3. Results

3.1. *Mining expansion trends*

Mining operations in Ghana, including different resources and types of operations (E.g. Large-scale industrial, small-scale artisanal etc.) expanded from 1 623 ha in 2000 to 53 391 ha in 2019, a gradual increase over my study period (Figure 2). Three broad expansion phases are visible in my time series data (See Fig. 2 below and Tab. 2 under appendix B). Between 2000 and 2010, mining activities expanded steadily, as they remained concentrated around established industrial mines at Tarkwa and Damang, with Kenyasi industrial mine leaving a significant impact on the numbers once operational in 2005. A significant acceleration then began in 2012, with annual additions of 4 035 ha (2012), and 5 674 ha (2013), representing a steep increase in land conversion to mining. This surge coincides with the peak of the global gold price and the widely documented influx of thousands of miners to Ghana's mining areas to take up mechanised artisanal mining. Growth then slowed temporarily in 2014 (2 369 ha) be-

fore resuming at an intensified pace from 2016 onward. 2017 through 2019, my study period's final years, stand out as statistically unusual, with z-scores of 1.69 (2017), 1.55 (2018) and 1.59 (2019), showing annual expansion rates well above the 20-year statistical average. These years collectively added over 20 667 ha of newly converted land within the mining areas.

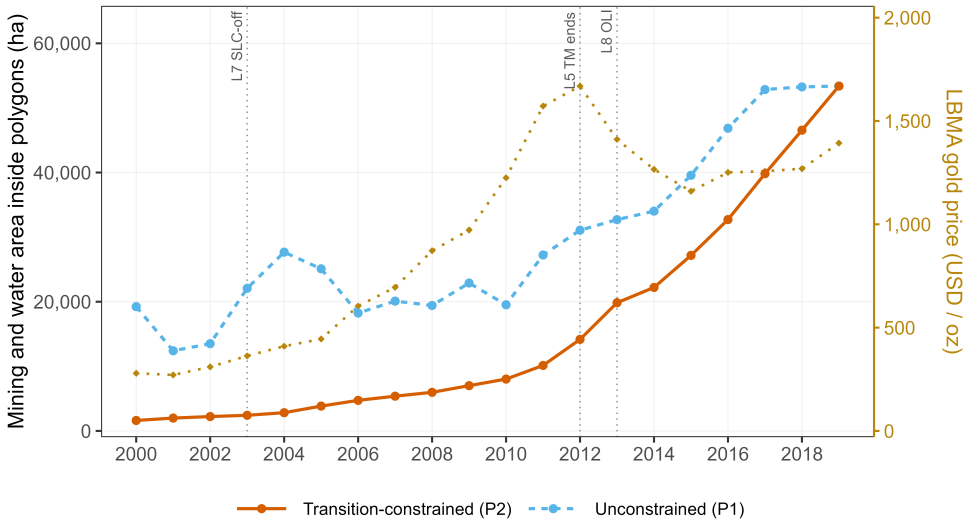


Figure 2. Comparison of unconstrained (blue) and constrained (red) classification results for 2000 to 2019. Gold price (USD) overlayed. Sensor eras indicated.

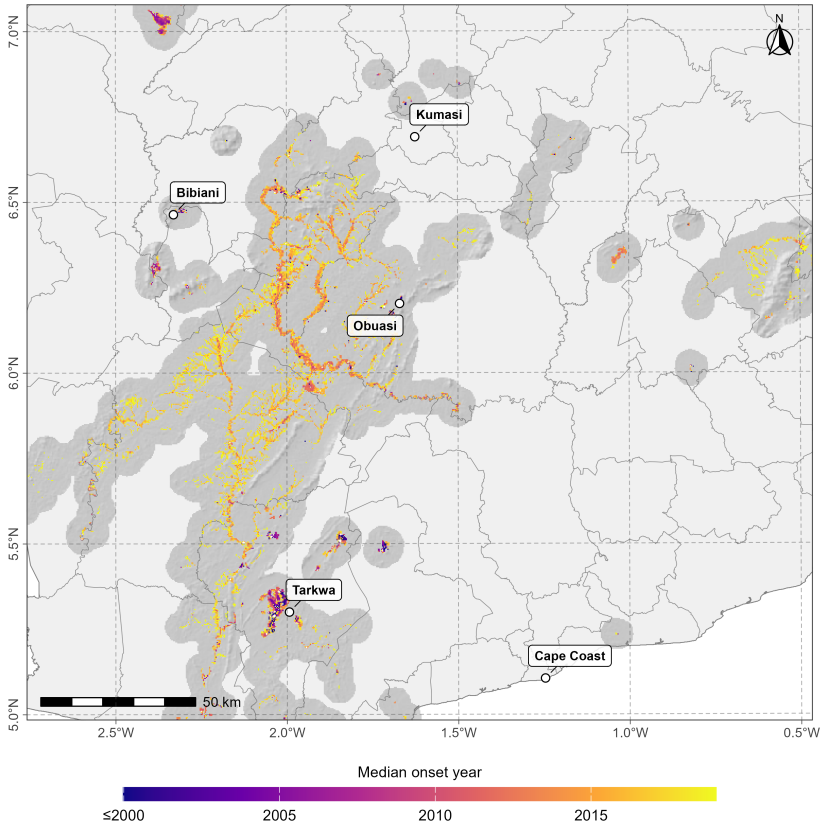


Figure 3. Mining onset across study area (Maus et al. (2022) polygons plus 5 km buffer).

Mining expansion was also concentrated geographically. Five districts make up the majority of Ghana's mining footprint: Tarkwa Nsuaem, Amansie West, Wassa Amenfi, Upper Denkyira, and Asutifi (Table 4, Table 5, Table 6, Table 7, Table 8 in Appendix.). Each of the districts shows unique expansion patterns reflecting local mining histories. For example Tarkwa Nsuaem is home to Ghana's largest open-pit gold operation and showed the most steady growth trajectory, expanding from 783 ha in 2000 to 8 149 ha in 2019. Expansion was gradual throughout the first decade, before accelerating sharply after 2016, with 2019 alone adding over 723 ha. This increase reflects not just mining intensification within established concessions, but also increased small-scale operations around industrial mining sites.

Amansie West experienced an abrupt single-year increase from 1 820 ha in 2012 to 3 854 ha in 2013 ($z = 2.14$), adding 2 033 ha in a single year likely due to the influx of up to 50,000 national and international miners (Aikins, 2024; Asiedu, Amaning, Koomson, Gyamfi, & Niang, 2023) and the beginning of the establishement of Asanko mine (not to be openend before 2016 though). In Wassa Amenfi, mining area remained below 1 000 ha until 2012, then grew rapidly to 11 657 ha by 2019. The years 2017 to 2019 showed the largest expansion of any district and year combination in this analysis ($z = 1.97$, 2.31, and 1.76), with annual additions exceeding 2 000 to 2 500 ha. This pattern is consistent with the rapid expansion of artisanal mining into previously forested concession areas in the Western Region (Asiedu et al., 2023; Asori et al., 2022), which now threaten food security, and environmental health (Barenblitt et al., 2021; A. K. Mensah et al., 2015; A. K. Mensah & Tuokuu, 2025).

Upper Denkyira followed a broadly similar path but at smaller scale, growing from

68 ha to 6 343 ha. Its earliest significant expansion occurred in 2013 ($z = 1.85$), with a hightened intensification visible after 2015. Lastly, Asutifi district stands out as its expansion was dominated by a single event in 2005 ($z = 2.55$), when the Newmont Ahafo mine became operational. The mining surface expanded from 13 ha to 381 ha (2005) and then 660 ha (2006) following the establishment of the mine before growth stabilised at a modest 107 ha/yr average for the remainder of the study period.

3.2. *Classification accuracy*

For my national model (2019), spatial cross-validation showed an overall accuracy of 0.87, with an overall kappa κ of 0.80 (Table 1 below). The model generally classified well across Ghana's diverse landscape, though a few things stand out. Mining detection in Ghana's North was challenging for the model (Mining F1=0.56), whilst water detection was best across the study area. This indicates an effect of the few available mining training points in the zone but also points to strong spectral contrast between water and vegetation in Ghana's north. The Forest Transition zone performed very well (0.88), whilst the Coastal zone (0.88) showed best seperation of the mining class (F1=0.83) despite being prone to coastal haze and heterogenous landscapes. Most confusion occurred between mining and vegetation, especially in the north, pointing towards the sparesly vegetated landscape there.

Table 1. 2019 national RF model performance under spatially blocked 5-fold cross-validation. Buffer distance of 2.2 km derived from empirical variogram (See section 3.5). Zone rows split by ecological zone; the model itself does not use zones.

Scope	n	Accuracy	95% CI	κ	Per-class F1		
					Water	Mining[b]	Veg.
National	**1 394**	**0.87 ± 0.03**	**[.83, .91]**	**0.80**	**0.93**	**0.78**	**0.88**
By latitude zone (same out-of-fold predictions)							
Northern Savannah[a]	361	0.88 ± 0.06	[.80, .95]	0.80	1.00	0.56	0.87
Forest-Transition	519	0.88 ± 0.07	[.79, .96]	0.80	0.92	0.82	0.89
Coastal	514	0.88 ± 0.05	[.82, .94]	0.81	0.89	0.83	0.90

Accuracies are mean ± SD across the 5 spatial folds; 95% CIs are t-distribution intervals across folds. F1 and Cohen's Kappa (κ) are fold means. Training data: 367 water, 391 mining and 636 vegetation samples (15 of 1 409 points dropped for missing band values). One-vs-rest AUC (national): water 0.99, mining 0.91, vegetation 0.95.

[a] Northern Savannah training data: 146 water, 46 mining and 169 vegetation samples. Only one mining site falls into the northern zone, so samples are few, making the zone estimate honest and more uncertain.

[b] Northern zone characteristics drag down overall F1 mining, but model predicts well in the major mining areas.

The achieved overall mining class F1-score (0.78) indicates good balance between precision and recall for the primary class of interest in most parts of my study area (South-Western Ghana (Forest transition and Coastal zones)). Systematic comparison of my national model vis-á-vis zone specific ones, with zone-specific features revealed that they perform similarily under identical conditions (See appendix 10 for full model performance assessment).

My results were also insensitive to the CV design with accuracy varying by less than 0.3 points for buffers of 0 to 10 km (the fitted variogram range is 0.2 to 6 km) and block sizes of 10 to 100 km. Random CV was only 0.7 to 0.9 points more optimistic, and the median test-to-training distance in my spatial CV (12.5 km) exceeded that of mapped

pixels inside the mining polygons (3.3 km)[6].

4. Discussion

4.1. *Comparison with existing estimates*

My analysis points towards 51 768 ha of converted land to mining between 2000 (1 623 ha) and 2019 (53 391 ha) within known mining polygons, following steep expansion trends after 2012, also observed by other studies (Barenblitt et al., 2021; Forkuor et al., 2020; Gallwey et al., 2020; Obodai et al., 2019; Snapir et al., 2017). Therein, my estimates sit between those of Barenblitt et al. (2021) and Gallwey et al. (2020), which both tried to mainly estimate land conversion to small-scale mining (but capture some industrial footprints). Whilst Barenblitt et al. (2021) attempt to map historical *artisanal* mining expansion in Ghana (2007-2019), reporting 47 414 ha of vegetation converted to mining in Ghana's southwestern forest zone, Gallwey et al. (2020, p.10), using a CNN on high-resolution imagery (Sentinel at 10m resolution) in recent years (2015-2019), show an increase of total converted land to small-scale mining from 64 000 ha (2015) to 77 000 ha (2019). Both studies rely on a study-, or "catchment," area in Ghana's south, creating estimates that exclude Ghana's north, as well as industrial operations and other minerals than gold, which are captured by the Maus et al. (2022) polygons I use.

Adding to that, my estimates include already operational, earlier mining, as well as small-scale operations outside of the study area of Barenblitt et al. (2021), Snapir et al. (2017), Forkuor et al. (2020), Obodai et al. (2019), and Gallwey et al. (2020), rendering my numbers more relevant for all of Ghana and over time (2000-2019). On overall classification accuracy, my model achieves 87% over 84% reported by Barenblitt et al. (2021). I also report spatially blocked cross-validation, uncertainty of the probabilistic estimator, and per-class F1 scores (harmonic mean between precision and recall or "producer's and user's accuracy'), missing from their validation (Barenblitt et al., 2021, p.5). In contrast, Gallwey et al. (2020, pp.10,12) show the superiority of deep learning approaches over random forest classifiers in more recent years where high resolution imagery (<10m) is available (3-6 % higher accuracy), but note that patch classifiers and U-Nets create overestimates by "smoothing," classifying forests, bare surfaces, and river systems (wet sediment and polluted rivers) as mining. Therefore, their estimate should be understood as an upper bound and showcases that deep learning approaches are impractical for estimating expansion in earlier years with coarser resolution imagery (30m Landsat era) as the classifiers require visual context.

Image quality and availability remains an issue for mining remote sensing, as Barenblitt et al. (2021) had to drop pre-2007 imagery for quality reasons, Forkuor et al. (2020) note that approximately 25% of Sentinel-1 scenes in their study were unusable due to atmospheric interference, and Gallwey et al. (2020) see temporary classification errors during the 2015-2016 El Niño event. Whilst Barenblitt et al. (2021) and Gallwey et al. (2020), amongst others (Forkuor et al., 2020; Obodai et al., 2019; Snapir et al., 2017), attempt to distinguish mining from urban land cover, leading to severe class confusion (partly due to data quality, partly due to classifier choices), especially along river systems with wet sediment, I opted for aggregating these classes within known mining polygons, capturing mining infrastructure expansion, too, which their approaches do not detect. Whilst I capture most small-scale and industrial mining,

[6]See also Appendix C2 and Fig. 13 for spatial assessment of cross-validation.

Barenblitt et al. (2021) and Gallwey et al. (2020) can capture smaller pockets of mining operations further away from the Maus et al. (2022) polygons but risk more class confusion between bare soil, urban, and actual mining pixels. My approach resolves this confusion by aggregating within the polygons.

Finally, Forkuor et al. (2020) note a slowing of "artisanal" mining expansion between 2015/16 and 2018/19 using Sentinel-1 backscatter thresholds, and Gallwey et al. (2020) suggest a 6.000 ha redcution in ASM expansion for 2017. My own, and others' estimates (Barenblitt et al., 2021; Obodai et al., 2019; Snapir et al., 2017) show the opposite trajectory: annual expansion accelerated from 5 546 ha in 2016 to 7 100 ha in 2017, with the final three years of my study period pointing to intensification. The divide may be explained by their applied change detection method which captures newly cleared surfaces but may undercount expansion within already-existing mining landscapes (Forkuor et al., 2020), their smaller study areas, data quality, and overcounting of the deep learning classifier (Gallwey et al., 2020).

These contrasting findings underscore that methodological choices regarding study period, classification area, class definitions, reference data, sensor type, change detection approach, and temporal constraints can substantially alter conclusions about mining trajectories, and assumptions about the effectiveness of policy interventions. Using the polygons of Maus et al. (2022), this study contributes a validated annual time series of mining footprints for Ghana from 2000 to 2019, taking temporal and spatial consistency into account, and enabling socio-economic assessments of mining expansions.

4.2. *Effects of temporal constraints*

Especially in the earlier years of my study period, the transition constrains removed a lot of mining area as compared to the unconstrained predictions (Table 2). This signals the reality of data availability, quality, and model uncertainty for these years. Without constraints, pixels could be classified as mining in 2005, vegetation in 2006, and mining again in 2007, which reflects an ecologically implausible sequence suggesting poor data quality and difficult classification rather than actual land cover dynamics (This has also been noted by Barenblitt et al. (2021, p. 9) and Forkuor et al. (2020, p. 2)). The observed mining expansion in earlier years of the study period corresponds well to qualitative evidence that suggests an intensification of mining activities in the years after new liberalisation laws passed in 2006 (Andrews, 2018; Asiedu et al., 2023; Asori et al., 2022), as well as concession records and production statistics (Ghana Minerals Commission, 2026a).

The unconstrained predictions show a significantly higher baseline of mining (19 244 ha in 2000) within validated polygons (Figure 2)[7] However, the unconctrained expansion trajectory is not linear and cautions against using these numbers: the unconstrained estimates fluctuate between consecutive years, even dropping from 27 678 ha in 2004 to 18 252 ha in 2006, and from 22 899 ha in 2009 to 19 531 ha in 2010, despite major mines opening during these years (E.g. Kenyasi (2005), Akyem (2013), Asanko (2016)), slow reclamation and long-lasting pollution of mining sites, increased industrialisation of small-scale mining, and an influx of foreign miners (Kumah, 2023; J. P. Mensah, Oduro-Kwarteng, Miezah, Boakye, & Koomson, 2025). The literature and qualitative evidence tell a story of increased investments in mining during these years, not less. Fluctuations in the unconstrained area predictions probably reflect

[7] See also Table 2.

poor data quality or genuine class confusion over genuine land-cover reversals (See data quality assessment in appendix D).

To further verify the transition constraints, I extracted class probabilities (N=1000 simulations) across the study period, summing per-pixel class probabilities for mining. This raises the estimated disturbed area inside the polygons from 53 391 to 68 061 ha in 2019 (and from 1 623 to 8 945 ha in 2000). This is because the many vegetation pixels (vegetation is the majority class throughout) each add a small probability of mining. Since my random forest was trained on a balanced, class-weighted sample (54% mining and water), its probabilities overstate disturbance where vegetation dominates (See probability-weighted curve in Figure 5). I therefore treat this probability-weighted estimate as an upper bound and report the ensemble-argmax numbers as my main results (Fig 5 and Table 3).

Lastly, I validated my approach of contraining earlier predictions with mining onset data from the Ghana Minerals Commission (Ghana Minerals Commission, 2026a, 2026b), as well as a trained CNN for major industrial mines (See section 3.7 and appendix Table 12)(Kattenborn et al., 2021; Ma et al., 2019; Nursamsi et al., 2024) and looked at mining onset at major small-scale and industrial operations (See Fig 15 and Fig 4). This allows me to visually interpret my classification rasters for the years of known mining onset and verify against the onset year identified by a CNN, representing deep learning approaches (See appendix Table 12, Figure 4 and Figure 3 below). As expected, the RF does a better job at classifying back in time to the noisy earlier landsat imagery (See Table 12 and Figure 14) but does not over-restrict pixel-transitions at the same time. Onset years of major industrial mines are captured well, whilst the CNN struggles in earlier years and sees an industrial mine where it only was developed years later. This confirms the insights of the current state of methods, where deep learning approaches rely on good training data and image quality in order to outperform other classification approaches (Kattenborn et al., 2021; Ma et al., 2019). RF classifiers on the other hand do well with few training points and perform consistently even in lower-quality data periods.

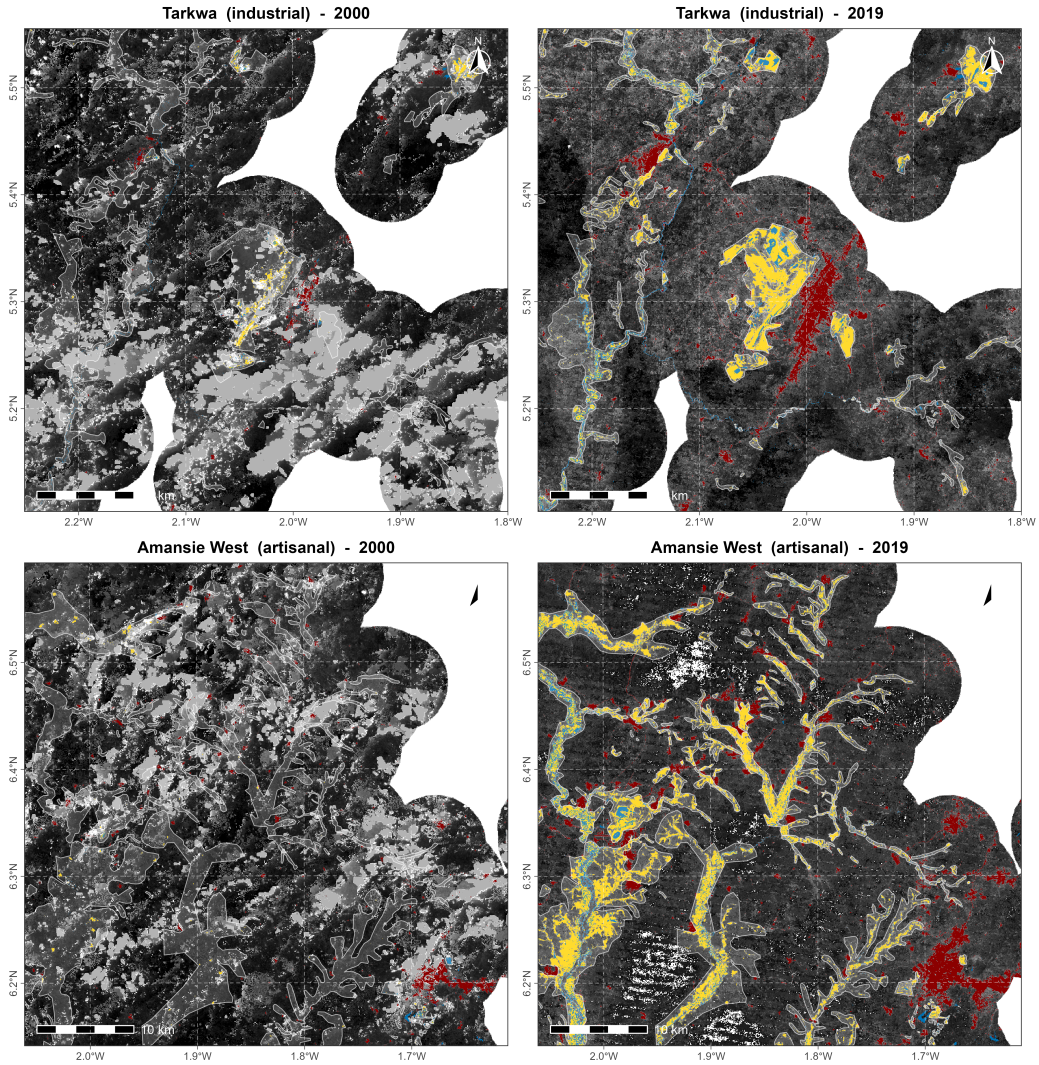


Figure 4. Mining expansion visible at Tarkwa mine and artisanal mining sites close to Obineben. Mining polygons overlayed in transparent white. Mining (water and mining) cells in yellow. NA (cloud pixels) greyed out. Dark red cells outside the mining polygons represent built-up and bare soil, such as towns, streets and other infrastructure.

4.3. *Limitations*

In my work I applied quantile-based clipping on my feature bands to constrain outliers (Preserving 99,5% of valid data), but some inter-annual data quality issues may persist. Composite data quality assessment revealed that early-period predictions (2000–2005) are built on substantially fewer cloud-free observations than later years (See appendix Figure 14). The 2000 composite averaged fewer than four valid pixels per location across all zones, limiting the medoid algorithm's ability to select a good observation and increasing susceptibility to residual contamination (clouds) and sensor noise. Predictions for these years should therefore be interpreted with greater uncertainty than post-2012 estimates, although mining onset was predicted quite reliably as compared to the trained CNN (See appendix Table 12). Landsat 8 availability then increased valid pixel counts to 15–30 per composite pixel. Despite these limitations, no composit-

ing method can overcome the fundamental scarcity of cloud-free Landsat observations over tropical West Africa before 2013. The applied medoid approach preserves inter-band spectral relationships better than alternative compositing strategies that risk the 'Frankenstein effect' of combining observations from different atmospheric conditions (Flood, 2013). As visible in Figure 13, my models predict well across Ghana, but are more uncertain for mining in the north due to the little training data and mining sites located in Ghana's north (Maus et al., 2022).

My work has focused on estimating the extent and timing of Ghana's mining expansion using remote sensing. Whilst apparently apolitically in nature, I want to echo scholars before me who have noted the 'top-down' nature of remote sensing analyses (Barenblitt et al., 2021, p.9), cautioning against using these methods as a tool to make decision about people without enganging in a deeper dialogue on what the observed changes mean for individuals living in Ghana, taking into account people's perspectives (Lund & Boone, 2013; Rutazibwa, 2023).

Whilst the polygons help me produce detailed estimates for Ghana's historical mining expansion at disaggregated level, mining activties outside the 2019 polygon boundaries cannot be assessed. Any mines that were active historically but abanoned before 2019, and some artisanal operations (E.g. under tree canopy) too small to appear in the dataset, can not be captured using my approach. I accept these limitations in my study as Ghana's mining boom is ongoing (Asiedu et al., 2023; Asori et al., 2022; Siachiwena, 2025), and it can be assumed that even if some mines were abandoned[8], people will either e.g. try to revisit sites with newer equipment, and/or sites won't be usable or transformed until after natural vegetation has regrown from the degraded land. For assessing national and local expansion trajectories within currently active mining areas (2019), my constrained approach is appropriate and produces results that can be linked to population or environmental data for further analysis of mining impacts.

4.4. *Implications for mining impact assessment*

By expanding the definition of the mining landcover class in my study, my approach provides meaningful impact estimates that improve on methods counting mining cells across larger study areas and risk severe class confusion in earlier Landsat years. Tailings ponds, processing facilities, roads, worker housing, and cleared land around mining sites all represent land cover changes with environmental and social consequences (Andrews, 2018; McQuilken & Hilson, 2016) which can be quantified using my suggested approach and the Maus et al. (2022) dataset. Previous studies may have systematically underestimated mining footprints by classifying such infrastructure as urban or bare soil distinct from mining (Barenblitt et al., 2021), or overestimated by risking class confusion between mining, urban, and bare soil (Snapir et al., 2017), or having CNN or U-Net smoothing over mining details in earlier years (Ball et al., 2017; Kattenborn et al., 2021; Ma et al., 2019). Eventhough my expansion estimates are slightly higher than e.g. the ones of Barenblitt et al. (2021), my numbers are likely underestimates, as areas outside of mining polygons Maus et al. (2022) were not included in this study (Gallwey et al., 2020).

The pixel-year based time series data generated by this analysis can support quantitative investigations of mining's relationship with demographic, economic, social, and environmental outcomes when linked to survey or administrative data. My data shows where mining extents expanded most severely (See Fig. 3 below), and corresponds well

[8] See for example J. P. Mensah et al. (2025) on soil remediation and health risks at an ASM site.

to ground-truth data on industrial mine establishment, and Ghana's artisanal boom from around 2011 (Ghana Minerals Commission, 2026a). The IPUMS-compatible district boundaries (Ruggles et al., 2025) facilitate integration with census microdata for further analyses. For Ghana specifically, my data allows the assessment of policy implications (Such as operation Vanguard targeting artisanal mining (Hilson, 2017)), and the study of mining impacts on local communities.

5. Conclusions

My analysis demonstrated that polygons of known mining sites can effectively anchor temporal analyses of mining expansion in Ghana. Constraining classification to known mining areas from recent high-resolution footprints, makes possible the assessment of historical mining-related land cover change. This method includes both industrial and "small-scale" operations, mining infrastructure, and tailings, contributing meaningful new estimates, especially in a historical perspective and for integration with survey data. My model, achieving overall accuracy of 0.87 shows good applicability across Ghana, vis á vis deep learning methods (CNN), and validated onset information from industry statistics aquired from the Ghana Minerals Commission.

Transition constraints prooved effective in preventing ecologically implausible class dynamics and produce more conservative land cover change estimates as compared to unconstrained classifications. I estimate mining expansion in Ghana from approximately 1 623 ha in 2000 to 53 391 ha in 2019, concentrated in southwestern Ghana's forest zone with peak expansion during 2012 to 2013, and from 2016 onward. These findings are in line with qualitative evidence and the establishment of major industrial mines in Ghana's districts.

For Ghana specifically, this study contributes knowledge about the *Where* and *When* of all mining activities from 2000 to 2019 on national and district level, combining industrial and "artisanal" for the first time in a time series from 2000-2019. My data shows when and where mining operations began and how they expanded over time. This information is relevant for local populations, researchers, and policymakers to assess how these quantifiable landcover changes coincide with policy shifts (E.g. Operation vanguard, GoldBod) and impacts on humans and the environment.

The methodological framework presented here, using validated polygons as focused classification area, zone-specific model validation, and transition constraints, might be transferable to other mining regions. The global coverage of the Maus et al. (2022) mining polygons suggests broad applicability for historical mining expansion analysis. Besides improving on estimates, these methodos have relevance for ecological realism, and related socio-economic analyses of global mining impacts.

Acknowledgements

I would like to thank Jonas Vestby for his continued support and expertise along the way. Critical comments and ideas by Tone Sommerfelt and the LANDRESPONSE team led by Tor A. Benjaminsen helped refine this article. I am also grateful for the input of the Skogrover lunch group at MINA (NMBU).

Disclosure statement

No potential conflict of interest was reported by the author.

Funding

This work was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme [LANDRESPONSE, grant agreement number 101054410].

Data availability statement

The annual mining footprints for 2000 to 2019, the training data and the area statistics are openly available in Zenodo at `https://doi.org/10.5281/zenodo.24014446`. The Google Earth Engine and R code used to produce them is archived in Zenodo at `https://doi.org/10.5281/zenodo.24013792`, with the development version at `https://github.com/hannesbra/Braeuer_26_ghana_mining_expansion`.

Landsat imagery was accessed through Google Earth Engine (Gorelick et al., 2017). Mining polygons are available from Maus et al. (2022). District boundaries can be accessed from Ruggles et al. (2025).

Use of generative AI

During the preparation of this work the author used Claude Opus 4.7 and 5.5 (Anthropic) to write and streamline Google Earth Engine and R code for producing the satellite composites and the machine-learning analyses. After using this tool, the author reviewed, tested and edited the code and its outputs as needed and takes full responsibility for the content of the publication.

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6. Ecological zone descriptions

As Ploton et al. (2020) and Zhang et al. (2026) show, large-scale ecological mapping models often overestimate their predictive power, and don't take into account locally varying spectral signatures. This is why I divided my study area into three ecological zones with distinct characteristics relevant to mining detection, following the agro-climatic reclassification of Yamba et al. (2023). I validate these ecological zones against my national RF model.

Training points were assigned to ecological zones based on latitude thresholds calibrated to balance ecological representativeness[9]. Zone boundaries were set at 8.0°N (northern/forest-transition) and 6.0°N (forest-transition/coastal) according to reclassification of Yamba et al. (2023).

Northern Savannah Zone (≥8.0°N): Ghana's north receives unimodal rainfall (ca. 800–1 200 mm annually) and is mostly sparsely vegetated. Only one mining site falls into the north. Sparse vegetation helps to detect bare-soil signatures but also increases spectral confusion between mining and vegetation classes. Literature-based zone features emphasise BSI and the SWIR bands.

Forest-Transition Zone (6.0 to 8.0°N): The forest zone has higher, bimodal rainfall (1 200 to 1 800 mm) and consists of dense forest, plantations and agricultural land. Mining comprises industrial open-pit and underground operations, and alluvial gold extraction along rivers. Dense vegetation gives strong contrast to mining, but persistent cloud

[9]See training data, and ecological zones in Figure 1.

cover and shadowing complicate compositing, especially before 2013. Literature-based zone features emphasise NDVI, MNDWI and texture metrics for detecting canopy disturbance.

Coastal Zone (<6.0°N): The coast has bimodal rainfall with coastal influences, including mangroves, coastal savannah and intensive cultivation. Mining includes beach-sand extraction and gold operations in river deltas. Water indices (MNDWI, AWEI) are expected to be important here.

7. Annual expansion statistics

Table 2. Estimated mining area (ha) inside known polygons (Maus et al., 2022), 2000 to 2019. Area combines mining and water class pixels within the polygons. Unconstrained area is the per-year argmax classification from the national random forest ensemble; constrained area applies the transition (shrinking) rule. Annual change and z-scores refer to the **constrained** estimates.

Year	Unconstrained (ha)	Constrained (ha)	Annual change (ha)	z-score[a]
2000	19 244	1 623	—	—
2001	12 410	1 993	370	−0.91
2002	13 511	2 227	233	−0.96
2003	22 081	2 424	198	−0.98
2004	27 678	2 825	400	−0.90
2005	25 102	3 845	1 020	−0.66
2006	18 252	4 736	890	−0.71
2007	20 099	5 381	645	−0.80
2008	19 421	5 981	600	−0.82
2009	22 899	7 012	1 031	−0.65
2010	19 531	8 028	1 016	−0.66
2011	27 236	10 139	2 111	−0.24
2012	31 078	14 174	4 035	0.51
2013	32 716	19 848	5 674	1.14
2014	34 027	22 216	2 369	−0.14
2015	39 575	27 178	4 961	0.86
2016	46 860	32 724	5 546	1.09
2017	**52 855**	**39 824**	**7 100**	**1.69**
2018	**53 263**	**46 560**	**6 736**	**1.55**
2019	**53 391**	**53 391**	**6 831**	**1.59**

[a] Years with unusually high expansion over the study period's mean ($z > 1.5$) are shown in **bold**.

7.1. *Uncertainty validation of the constrained approach*

Table 3. Simulation of disturbed area (mining + water) within polygons, 2000 to 2019 ($n = 1\,000$ draws). In each draw one of the five ensemble members is selected at random for every year; the most likely class is mapped and the transition rule is applied. Deterministic = constrained map of the averaged ensemble. PI = empirical prediction interval (2.5th and 97.5th percentiles across draws); relative width = PI width / mean. All areas in hectares.

Year	Deterministic	Simulation mean	95% PI	95% PI width	Relative width (%)
2000	1 623	1 424	1 332–1 560	228	16.0
2001	1 993	1 767	1 651–1 924	273	15.4
2002	2 226	1 998	1 867–2 171	304	15.2
2003	2 424	2 200	2 050–2 379	330	15.0
2004	2 825	2 599	2 399–2 841	442	17.0
2005	3 845	3 601	3 335–3 951	616	17.1
2006	4 736	4 462	4 200–4 790	591	13.2
2007	5 381	5 090	4 830–5 421	591	11.6
2008	5 981	5 715	5 406–6 073	667	11.7
2009	7 012	6 695	6 380–7 051	671	10.0
2010	8 028	7 700	7 366–8 125	759	9.9
2011	10 138	9 803	9 326–10 378	1 052	10.7
2012	14 174	13 744	13 043–14 637	1 594	11.6
2013	19 848	19 193	18 320–20 294	1 974	10.3
2014	22 216	21 537	20 512–22 690	2 178	10.1
2015	27 178	26 425	25 129–27 951	2 823	10.7
2016	32 724	31 984	30 307–34 121	3 814	11.9
2017	39 824	39 062	36 869–41 713	4 844	12.4
2018	46 560	45 888	43 146–50 307	7 161	15.6
2019	53 392	53 732	50 263–57 163	6 900	12.8

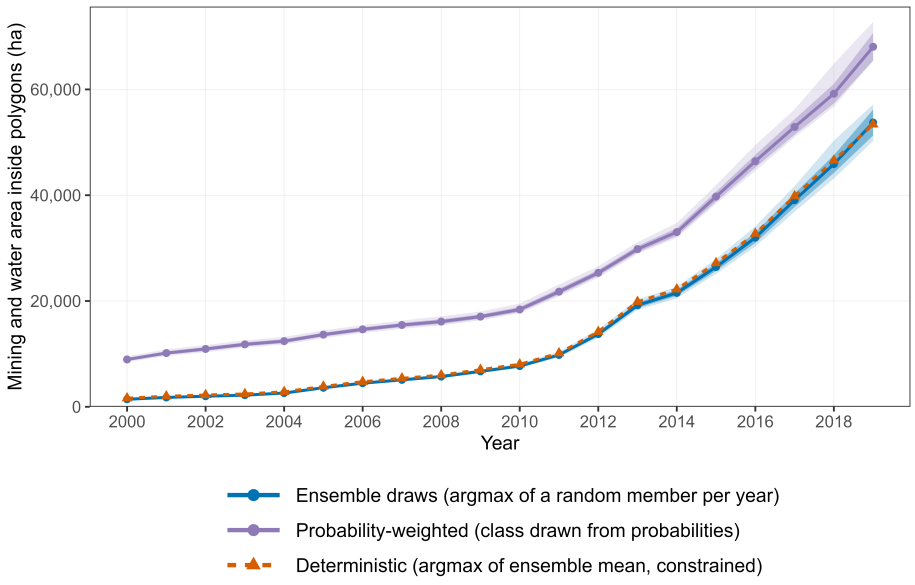


Figure 5. Deterministic (argmax of the ensemble mean, then constrained) versus ensemble draws (argmax of a randomly drawn member per year) versus probability-weighted draws (class sampled from the member probabilities), disturbed area within polygons. Shaded bands show ±1 SD (dark) and the 95% interval (light).

7.2. *District mining expansion trends*

Table 4. Estimated mining area within polygons (mining + water), **Tarkwa Nsuaem** (the merged Tarkwa Nsuaem, Prestea and Huni Valley districts), 2000 to 2019. The constrained estimate applies the transition rule. z-scores are computed against the district mean annual change ($\bar{x} = 388$ ha, $s = 191$ ha). Years with $z > 1.5$ in **bold**.

Year	Unconstr. (ha)	Constr. (ha)	Annual change (ha)	z-score
2000	4 239	783	—	—
2001	3 750	1 012	230	−0.83
2002	3 990	1 169	157	−1.20
2003	4 769	1 291	122	−1.39
2004	6 522	1 555	263	−0.65
2005	6 563	1 988	433	0.24
2006	6 198	2 298	310	−0.40
2007	6 698	2 503	204	−0.96
2008	6 751	2 695	192	−1.02
2009	7 382	3 086	391	0.02
2010	7 253	3 364	278	−0.57
2011	8 901	3 865	502	0.60
2012	8 388	4 409	544	0.82
2013	7 178	4 844	435	0.25
2014	6 977	5 110	266	−0.64
2015	7 266	5 509	399	0.06
2016	7 664	5 972	463	0.39
2017	**8 203**	**6 708**	**736**	**1.82**
2018	**8 150**	**7 426**	**718**	**1.73**
2019	**8 149**	**8 149**	**723**	**1.75**

Table 5. Estimated mining area within polygons (mining + water), **Amansie West** (Amansie West district), 2000 to 2019. The constrained estimate applies the transition rule. z-scores are computed against the district mean annual change ($\bar{x} = 580$ ha, $s = 678$ ha). Years with $z > 1.5$ in **bold**.

Year	Unconstr. (ha)	Constr. (ha)	Annual change (ha)	z-score
2000	3 877	150	—	—
2001	2 194	183	32	−0.81
2002	2 271	199	17	−0.83
2003	3 803	211	11	−0.84
2004	4 955	221	11	−0.84
2005	3 162	244	23	−0.82
2006	1 474	284	40	−0.80
2007	1 676	302	18	−0.83
2008	1 842	338	35	−0.80
2009	2 911	460	122	−0.68
2010	1 463	607	148	−0.64
2011	2 228	862	255	−0.48
2012	4 096	1 820	958	0.56
2013	**6 331**	**3 854**	**2 033**	**2.14**
2014	6 797	4 487	633	0.08
2015	8 502	6 044	1 557	1.44
2016	10 189	7 468	1 425	1.24
2017	11 182	8 935	1 466	1.31
2018	10 943	10 003	1 068	0.72
2019	11 178	11 178	1 176	0.88

Table 6. Estimated mining area within polygons (mining + water), **Wassa Amenfi** (the merged Wassa Amenfi East and West districts), 2000 to 2019. The constrained estimate applies the transition rule. z-scores are computed against the district mean annual change ($\bar{x} = 610$ ha, $s = 896$ ha). Years with $z > 1.5$ in **bold**.

Year	Unconstr. (ha)	Constr. (ha)	Annual change (ha)	z-score
2000	3 156	76	—	—
2001	1 397	82	7	−0.67
2002	1 950	86	3	−0.68
2003	5 518	94	8	−0.67
2004	6 827	101	6	−0.67
2005	5 730	114	14	−0.67
2006	3 345	134	20	−0.66
2007	3 972	158	23	−0.65
2008	2 910	196	38	−0.64
2009	3 140	304	108	−0.56
2010	2 796	408	104	−0.57
2011	4 930	599	191	−0.47
2012	4 305	1 000	402	−0.23
2013	3 190	1 509	508	−0.11
2014	3 558	1 920	411	−0.22
2015	5 066	2 930	1 010	0.45
2016	7 663	4 408	1 477	0.97
2017	**10 563**	**6 785**	**2 377**	**1.97**
2018	**11 770**	**9 467**	**2 682**	**2.31**
2019	**11 657**	**11 657**	**2 190**	**1.76**

Table 7. Estimated mining area within polygons (mining + water), **Upper Denkyira** (the merged Upper Denkyira East and West districts), 2000 to 2019. The constrained estimate applies the transition rule. z-scores are computed against the district mean annual change ($\bar{x} = 330$ ha, $s = 346$ ha). Years with $z > 1.5$ in **bold**.

Year	Unconstr. (ha)	Constr. (ha)	Annual change (ha)	z-score
2000	1 251	68	—	—
2001	780	78	10	−0.93
2002	1 045	81	3	−0.94
2003	2 276	86	5	−0.94
2004	2 466	87	1	−0.95
2005	1 506	99	13	−0.92
2006	696	121	22	−0.89
2007	1 116	136	14	−0.91
2008	1 107	160	24	−0.88
2009	1 786	270	110	−0.64
2010	1 224	464	194	−0.39
2011	2 377	918	454	0.36
2012	3 201	1 700	781	1.30
2013	**3 803**	**2 669**	**969**	**1.85**
2014	3 961	2 934	265	−0.19
2015	4 721	3 556	622	0.84
2016	5 725	4 277	721	1.13
2017	6 422	5 078	801	1.36
2018	6 392	5 752	673	0.99
2019	6 343	6 343	591	0.75

Table 8. Estimated mining area within polygons (mining + water), **Asutifi** (Asutifi district), 2000 to 2019. The constrained estimate applies the transition rule. z-scores are computed against the district mean annual change ($\bar{x} = 107$ ha, $s = 102$ ha). Years with $z > 1.5$ in **bold**.

Year	Unconstr. (ha)	Constr. (ha)	Annual change (ha)	z-score
2000	700	1	—	—
2001	141	1	0	−1.05
2002	8	2	1	−1.04
2003	11	3	1	−1.04
2004	28	13	11	−0.95
2005	**693**	**381**	**368**	**2.55**
2006	**985**	**660**	**279**	**1.68**
2007	1 274	912	252	1.42
2008	1 575	1 114	203	0.93
2009	1 580	1 210	96	−0.11
2010	1 596	1 304	94	−0.13
2011	1 754	1 444	140	0.32
2012	1 881	1 543	99	−0.08
2013	1 996	1 659	116	0.09
2014	2 145	1 771	112	0.04
2015	2 107	1 800	29	−0.77
2016	2 141	1 860	60	−0.46
2017	2 134	1 918	58	−0.48
2018	2 123	1 968	50	−0.56
2019	2 039	2 039	71	−0.35

8. Variable selection and model comparison

Recently, attention has been pointed towards local ecological realities and implications for effective land cover classification across heterogenous landscapes (Phiri & Morgenroth, 2017; Shen, Wang, Su, He, & Li, 2022; Zhang et al., 2026). Therefore, I investigated how my national model performs against ecological zone specific models, in different areas of Ghana, and how zone-specific features as informed by literature perform against data-driven feature selection (Chen, Vaudour, Richer-de-Forges, & Arrouays, 2025) (See appendix section C). In my study, where I modelled Ghana's three ecological zones after Yamba et al. (2023), I find that zone-specific models were statistically indistinguishable from a national model. However, the comparison revealed some meaningful nuances. Whilst a national model handles local signature differences implicitly (Larger feature set and sample size), different features impacted mining F1 in different parts of Ghana. For example, whilst the Forest-Transition zone showed good detection accuracy across classes (Mining F1 = 0.82; Water F1 = 0.92; Vegetation F1 = 0.89), the Northern zone showed more variability with a lower mining F1 of 0.56 and perfect water detection of F1 = 1.0. The feature importance test reveals that bare soil signatures play an important role, with BSI, CLAY and SWIR features dominating mining detection in the coastal zone, whilst BSI and NDBI, as well as NIR texture are important features for detection in the North. Overall, vegetation and mining were most often confused by the model, indicating the class confusion in the less-vegetated North, as well as confusion between shrubland and small mines in other zones.

Generally, this indicates that literature informed ecological zones matter and that a combination of data- and literature-informed, zone-specific variables (E.g. emphasising NDVI, SWIR, and texture metrics) can capture real ecological differences associated with local mining characteristics. The fact, that zone-specific feature sets, and zone-specific models performed as well as a national model calls for the importance of these features and classification areas grounded in ecology, as they are more interpretable.

A conducted power analysis (Buda et al., 2018) indicates that the national model is defensible for my use case but observed zone-specific performances suggest attention must be placed on ecological realities. Therein, my approach suggests that future work must take into account the unique spectral variations found across diverse ecological landscapes. Using verified feature sets for different ecological zones can help us improve classification accuracy and interpretability.

The mediocre performance of my northern zone on the other hand is rooted in the fact that only one small mining site (near Tinga in Savannah region) of the Maus et al. (2020) polygons falls into these latitudes, resulting in few, clustered mining samples. If future studies seek to focus on mining detection in Ghana's north, I suggest training separate models for this zone and type of vegetation.

8.1. *Feature selection*

The 2010 composite (example) was created from five to ten valid dry season pictures across 2009, 2010, and 2011, leaving the best medoid scene-images for the final composite creation. This approach follows recommendations by Flood (2013) who demonstrated that medoid-based selection preserves spectral relationships between bands better than simple median or maximum NDVI compositing. I excluded scenes that showed more than 50% cloud cover.

Landsat 5 TM and Landsat 7 ETM+ were used for composites from 2000 through

2012, transitioning to Landsat 7, and 8 OLI for subsequent years. Cross-calibration between sensors followed Roy et al. (2016) and USGS Collection 2 Level-2 surface reflectance specifications (USGS, 2022). I addressed Landsat 7 scan-line corrector failures by filling the gaps with overlapping observations from adjacent dates and sensors within each three-year window. Cloud masking utilized the QA_PIXEL band[10].

I included surface reflactance in blue, green, and red, as well as near infrared (NIR) and short-wave infrared (SWIR), which provide good information for mineral detection (Koerting et al., 2024). Several spectral indices were then derived from the original bands, enabling better discrimination based on reflectance values and ecological knowledge. The Normalized Difference Vegetation Index (NDVI; Rouse, Haas, & Deering, 1973; Tucker, 1979) was calculated for vegetation discrimination, whilst the Bare Soil Index (BSI; Rikimaru, Roy, & Miyatake, 2002) and modified BSI (Nguyen, Chidthaisong, Kieu Diem, & Huo, 2021) were used for soil detection. Water indices used include NDWI, Modified NDWI (MNDWI; Xu, 2006), and Automated Water Extraction Index (AWEI; Feyisa, Meilby, Fensholt, & Proud, 2014) for tailing pond detection, and mining along river systems. An Urban Index (UI; Javed et al., 2021) was employed alongside iron oxide ratios (red/blue) and clay indices (SWIR1/SWIR2) as potential indicators of mineralization (Barenblitt et al., 2021).

Adding to these, I included topographic information such as elevation, slope, aspect, and the Terrain Ruggedness Index (TRI) derived from SRTM 30 m data (Barenblitt et al., 2021; Farr et al., 2007; Svoboda et al., 2022). Additionally, I used grey-level co-occurrence matrix (GLCM) features such as contrast, correlation, variance, and entropy, calculated from the NIR band at 3×3 pixel windows (Haralick, Shanmugam, & Dinstein, 1973) to leverage local variance and higher contrast at mining sites (E.g. pits, stockpiles, roads) as compared to natural bare surfaces.

Lastly, I employed temporal metrics such as standard deviation of NIR, red, and SWIR1 bands across the three-year compositing window to capture intra-period variability potentially associated with active mining expansion. An NDVI anomaly layer was calculated as the difference between each year's NDVI and a baseline mean NDVI (Permutation-based from 2000 to 2004), providing a vegetation disturbance indicator relative to pre-mining conditions (Barenblitt et al., 2021). A full list of variables found in each composite, as well as their value ranges is shown below.

I also tested zone-specific features as I assumed local spectral signatures, vegetation, and mining types differ, demanding consideration of varying variables to include for model training. I balanced correlation of variables (For example water index family)(Hooker, Mentch, & Zhou, 2021), statistical significance of inclusion of a variable, and ecological knowledge from literature. I report standard permutation (Just to showcase the correlation of indices as pointed out by e.g. Hooker et al. (2021)), clustered permutation (Leaving out entire groups of variables when they are highly correlated such as e.g. the water indices), and a 'per feature, per zone, per fold' selection leaving individual variables, and their correlated 'clusters' out to establish their importance for classification. The top-k features of this data-driven selection are then tested using spatial clustering against (and combining) literature-derived feature selection using the McNemar (1947) test to determine whether one feature-set is significantly better at discriminating land cover. The validation showed no statistically significant improvement of using zone-specific feature sets and models (As will be discussed later). Therefore, all features were used to train the national RF classifier.

[10]For more information on data quality and availability see Appendix D.

Table 9. The 27 bands of the national random forest. All bands are computed from annual Landsat medoid composites (2000 to 2019) and SRTM. Model input range: values are clipped to these limits before training and prediction. Not used: entropy (constant in the 2019 training composite), nightlight bands (sensor change from DMSP to VIIRS in 2013) and composite quality bands.

Category	Variable	Description	Model input range
Spectral	BLUE	Blue surface reflectance	[0.000, 0.250]
	GREEN	Green surface reflectance	[0.000, 0.250]
	RED	Red surface reflectance	[0.000, 0.350]
	NIR	Near-infrared surface reflectance	[0.000, 0.750]
	SWIR1	Short-wave infrared 1 reflectance	[0.000, 0.750]
	SWIR2	Short-wave infrared 2 reflectance	[0.000, 0.700]
Spectral indices	NDVI	Normalised difference vegetation index	[−1.000, 1.000]
	NDWI	Normalised difference water index (green, NIR)	[−1.000, 1.000]
	MNDWI	Modified NDWI (green, SWIR1)	[−1.000, 1.000]
	AWEI_NSH	Automated water extraction index, no shadow	[−10.0, 5.000]
	AWEI_SH	Automated water extraction index, shadow	[−10.0, 5.000]
	BSI	Bare soil index	[−1.000, 1.000]
	NDBI	Normalised difference built-up index	[−1.000, 1.000]
	IRON_OXIDE	Iron oxide ratio (red / blue)	[0.000, 3.000]
	CLAY_INDEX	Clay mineral index (SWIR1, SWIR2)	[−1.000, 1.000]
	CI	Colouration index (red, green)	[−1.000, 1.000]
	BI	Brightness index (visible bands)	[0.000, 1.000]
	WATER_PROB	Water probability score	[0.000, 3.000]
Texture	NIR_STD_DEV	Local standard deviation of NIR	[0.000, 0.250]
	NIR_contrast	GLCM contrast of NIR	unclipped
	NIR_corr	GLCM correlation of NIR	unclipped
	NIR_var	GLCM variance of NIR	unclipped
Topographic	ELEVATION	Elevation, SRTM (m)	[−32, 850]
	SLOPE	Terrain slope (degrees)	[0.0, 75.0]
	ASPECT	Terrain aspect (degrees)	[0, 360]
	TRI	Terrain ruggedness index	[0.0, 85.0]
Temporal	NDVI_ANOMALY	NDVI deviation from 2000–2004 baseline	[−1.500, 1.500]

A permutation analysis of related features (See below) showed that the national model's skill is spread across correlated band groups. Dropping any single feature "family" changed spatial-CV accuracy by at most 0.6 percentage points and mining F_1 by at most 1 point, well within the fold-to-fold variation (SD = 3 points). Band selection did not help either. A selection of my top 10 bands (0.858) and the 19 literature-based bands (0.859) both performed below the full 27-band model (0.870; McNemar $p = 0.034$ against the top-10 set). The literature-based and data-driven rankings overlapped only partly (Jaccard 0.32–0.50). Within the ecological zones, bare-soil indices (BSI, Clay Index) were the most influential feature group for mining F_1 in the coastal zone, and topography mattered in the forest-transition zone (log-loss +0.07 when removed). In the Northern Savannah zone, with only 46 mining samples around one site (6 to 16 per fold), feature "family" effects on mining F_1 (±4 points) were indistinguishable from fold noise (SD 13–18 points) and only the spectral water/vegetation family affected accuracy (log-loss 0.22 → 0.34). The zone-specific model did classify the north more accurately (0.909 vs. 0.859; McNemar $p = 0.005$), but equally well with literature or all 27 bands, and without improving the mining F_1 (+0.02, 95% CI −0.13 to +0.16). I therefore use one national model with all bands and treat northern mining estimates as the least certain.

Figure 8 shows drop-family importance: each correlated group of bands (e.g. the

water indices) is removed in turn and the model is re-trained under spatial cross-validation. Figure 6 compares literature-based and data-driven band sets per zone. Figure 7 shows mean accuracy across the five spatial folds with 95% confidence intervals. All 27 candidate bands performed best and were used for the final model.

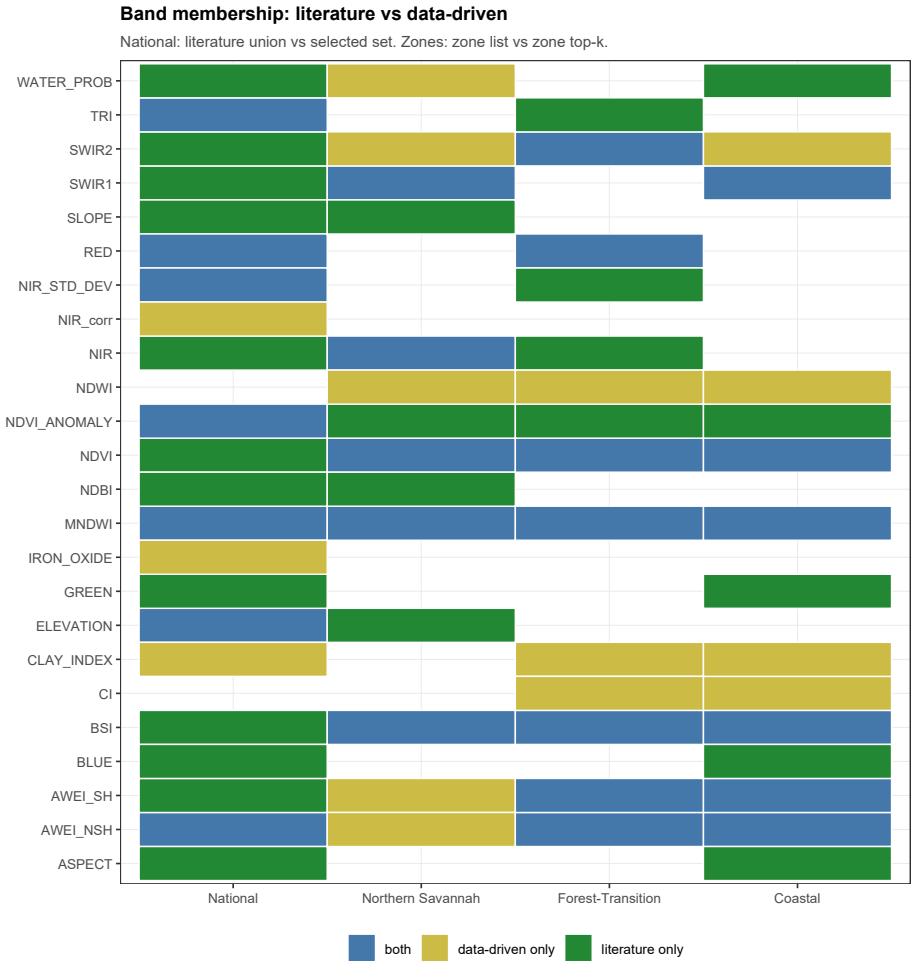


Figure 6. Band membership of literature-based and data-driven feature sets, nationally and per latitude zone.

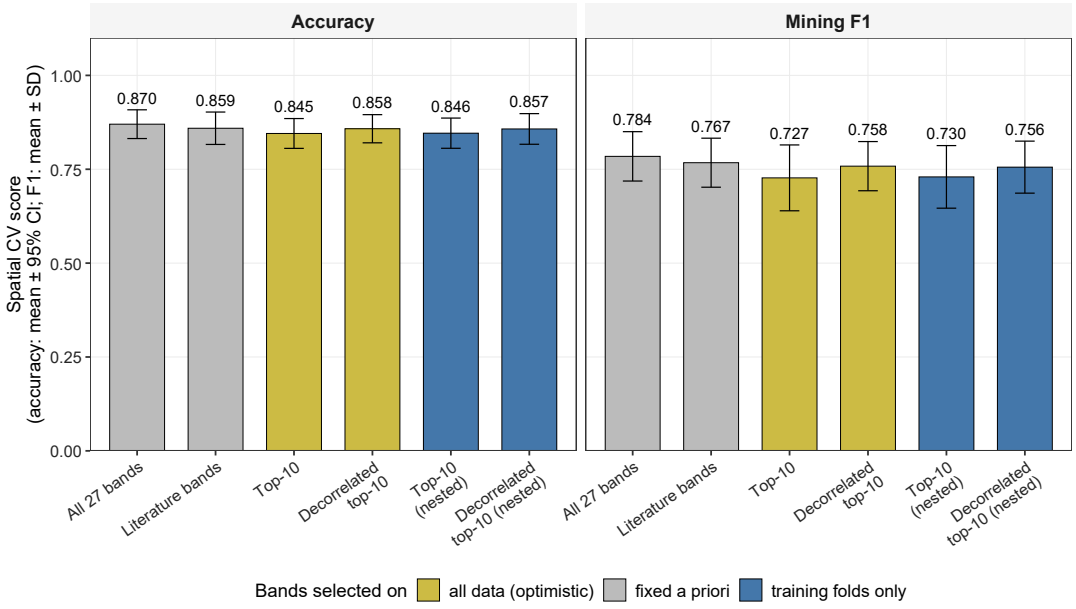


Figure 7. National band sets under spatial cross-validation (shared folds).

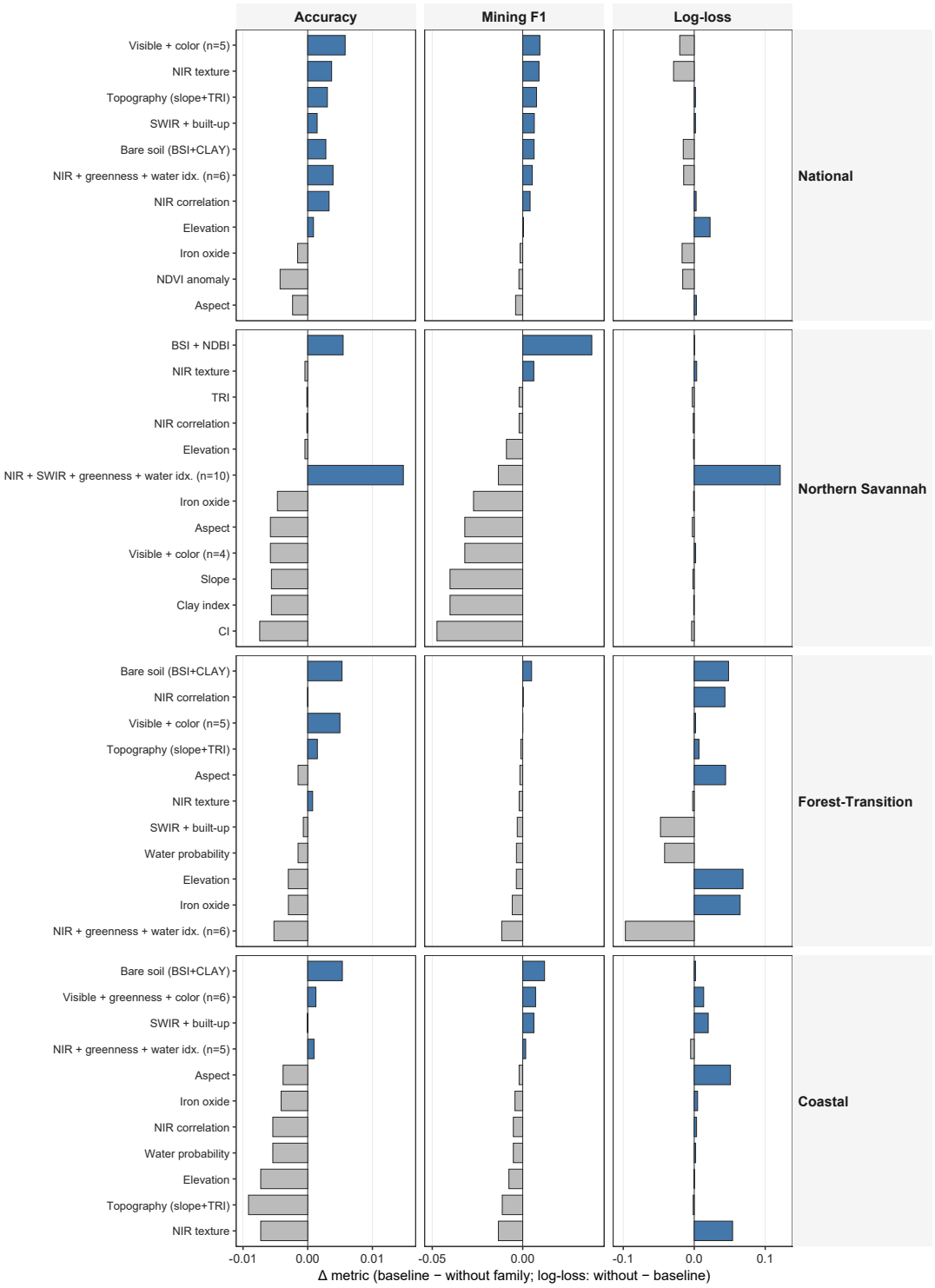


Figure 8. Effect of removing each band family on mining F_1, accuracy and log-loss.

8.2. *Model performance under spatial cross-validation*

Standard k-fold cross-validation overestimates accuracy by up to 50% when training and validation samples are spatially proximate (Roberts et al., 2017; Valavi, Elith,

Lahoz-Monfort, & Guillera-Arroita, 2019). Since my data is naturally clustered around mining sites and along river systems, I implemented spatial blocking cross-validation with five folds, using the `blockCV` package (Valavi et al., 2019) to divide my training and test data into spatially separate blocks within each fold (Meyer, Reudenbach, Hengl, Katurji, & Nauss, 2018). Training samples falling within the buffer around test blocks were excluded from that fold's training set. This approach provides more conservative and realistic accuracy estimates compared to random cross-validation (Ploton et al., 2020; Valavi et al., 2019). Based on my training data location I then conducted an 'area of applicability' analysis as advocated for by Meyer and Pebesma (2021) to assess the distance between points necessary to achieve 'spatial independence' and to compute block sizes appropriate for my data. These variograms assess the change in similarity between observations as distance between them is increased, resulting in a range needed to find data points that are uncorrelated.

As visible below in Figure 9, these variograms computed buffer sizes of 2,2 km. This range is used as buffer distance in the 5-fold cross-validation.

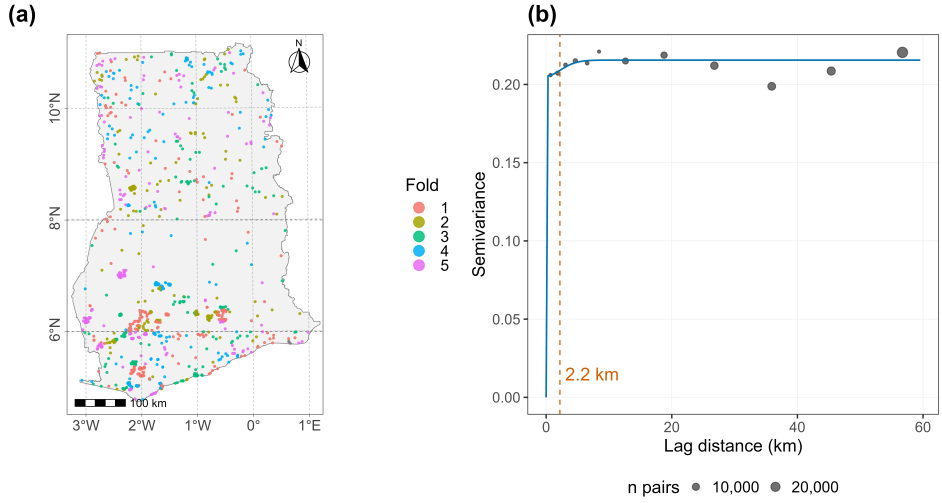


Figure 9. Spatial blocks (a) and Variogram fits (b) for spatial CV.

To evaluate whether ecological zoning and zone-specific feature selection provide measurable improvements over national or uniform feature approaches, I conducted a systematic comparison using the the same spatial folds and 22.5 km blocks[11] across the following configurations (See table 10 below). Three alternative combinations beside my national model (P0 - single national random forest model trained on all 27 features) were tested: (M1) a national model trained on only the 10 most important features as per the feature importance run; (M2) zone-stratified models using only literature-informed bands according to ecological knowledge; and (M3) zone-stratified models with all 27 bands. This setup ensures that observed differences reflect modelling choices rather than fold assignment. A McNemar's test was applied to out-of-fold predictions to assess whether my proposed model was significantly better at predicting land-cover classes (McNemar, 1947). A fourth configuration re-ran the proposed approach under random (non-spatial) cross-validation to quantify optimism bias from spatial autocorrelation (Meyer & Pebesma, 2021; Valavi et al., 2019).

[11] Compromise between the different zone-specific buffer distances for comparability.

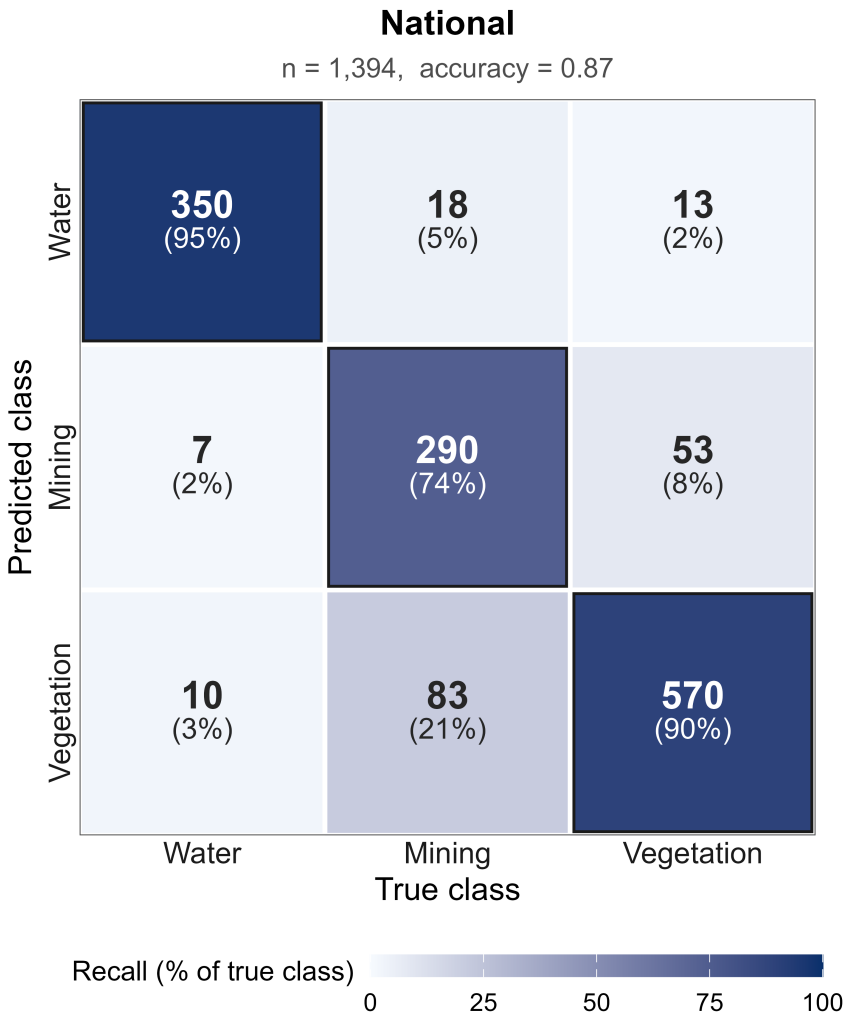


Figure 10. Confusion matrix of the national model.

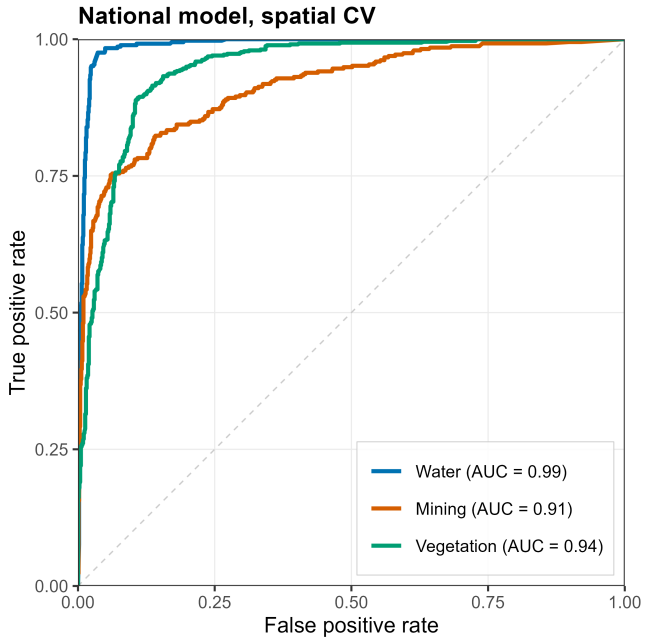


Figure 11. ROC curves of the national model (out-of-fold predictions, spatial cross-validation).

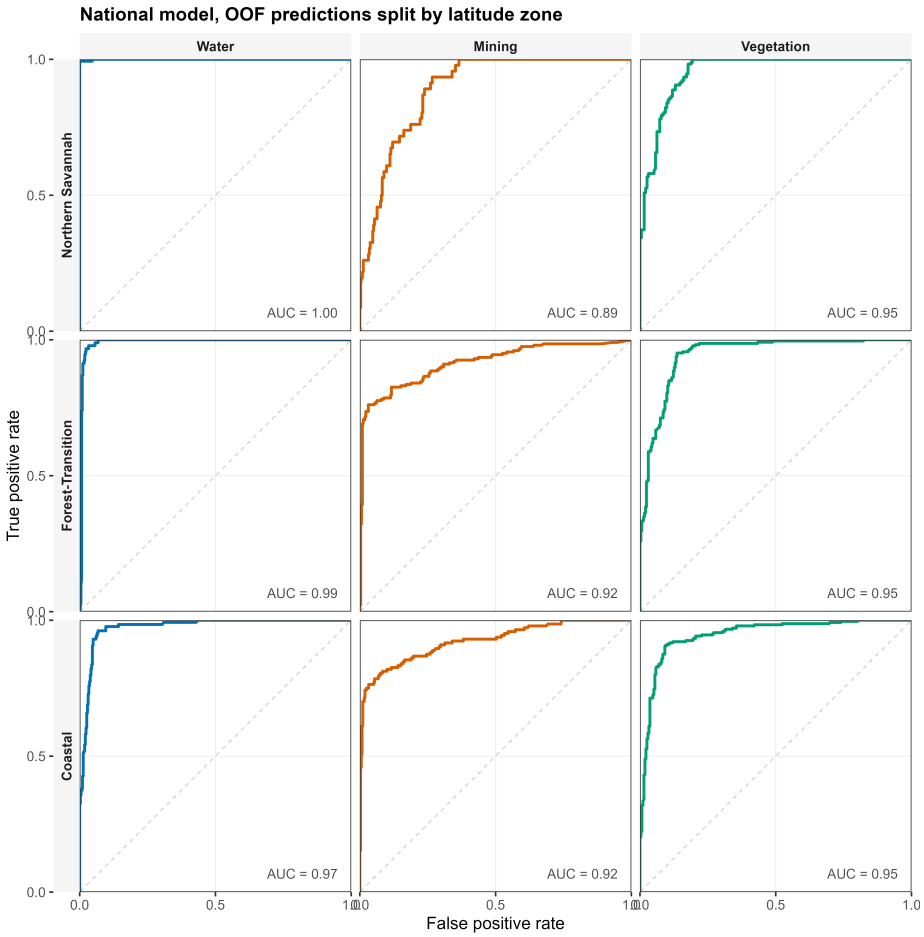


Figure 12. ROC curves of the national model per latitude zone.

The suggested national model (P0) achieved 0.87 (CI 0.83-0.91) overall accuracy (Mining F1 = 0.78) as compared to the national but 10-feature-focused model (M1) 0.858 (CI 0.82-0.9) (Mining F1 = 0.76), the zone-stratified and feature-focused model (M2) 0.87 (CI 0.86 to 0.88) (Mining F1 = 0.78), and the zone-specific but feature-uniform model (M3) 0.875 (CI 0.85-0.9) (Mining F1 = 0.79), with broadly overlapping 95 percent confidence intervals across all metrics (See Table 10 below). McNemar's tests indicate no statistically significant difference between my proposed national model and the zone-specific models but show that the full feature set (27 bands is significantly better than the national model with only 10 bands). These results suggest that statistical power was sufficient to determine that for my 1394 training samples distributed across three zones, there is no significant statistical difference between the national and zone-specific modelling approaches.

Random cross-validation inflated reported accuracy by only one percentage point (0.87 vs. 0.88) as compared to spatial blocking CV (Ploton et al., 2020; Valavi et al., 2019). This confirms that spatially proximate training and test samples share autocorrelated spectral signatures, and that the spatial CV estimates reported throughout this paper provide a more conservative and realistic assessment of model performances (Roberts et al., 2017; Valavi et al., 2019), but are also not too far off in my use case and show only a minimal pessimism bias.

Table 10 reports all metrics across modelling configurations using shared 5-fold spatial cross-validation folds with a 2.2 km buffer.

Table 10. Comparison of modelling approaches. Means from 5-fold spatially blocked cross-validation (22.5 km blocks, 2.2 km variogram-derived buffer) on $n = 1394$ training points; all spatial approaches share the same folds, so test points are matched (a prerequisite for the McNemar tests). 95% CI for accuracy from the t-distribution across folds. Random CV repeats P0 with random folds of identical size and no buffer; the gap to P0 shows the optimism of non-spatial validation.

Metric	P0: National (27 bands)[a]	M1: National (top-10 bands)	M2: Zone (literature bands)	M3: Zone (27 bands)	Random CV (P0)
Overall accuracy	0.870 [0.832–0.908]	0.858 [0.820–0.895]	0.869 [0.857–0.880]	0.875 [0.850–0.900]	0.879 [0.867–0.891]
Cohen's κ	0.796	0.777	0.794	0.803	0.810
Water F_1	0.935	0.942	0.923	0.931	0.942
Mining F_1	0.784	0.758	0.779	0.789	0.797
Vegetation F_1	0.879	0.865	0.885	0.887	0.887
Log-loss	0.405	0.434	0.441	0.376	0.342
McNemar p vs P0	—	0.034	1.000	0.396	—

[a] Model used for all maps. M1 uses the decorrelated top-10 bands from the feature audit; M2 is the zone-specific approach with literature-based bands per latitude zone; M3 uses zone models with the same 27 bands as P0, separating the effect of zoning from the effect of band choice.

To assess where the model's cross-validated performance can be expected to hold, I computed the dissimilarity index (DI) and a resulting area of applicability (AOA) following Meyer and Pebesma (2021) (Figure 13). The DI was generally low across the detected mining area, indicating that my prediction locations were similar to the training data and that the spatial CV error estimates are likely transferable to most of the map. Higher DI values were spatially clustered rather than random, concentrating along river systems and around industrial mining sites, where conditions such as wet sediments or sparse vegetation regrowing on bare soil were underrepresented in the training data.

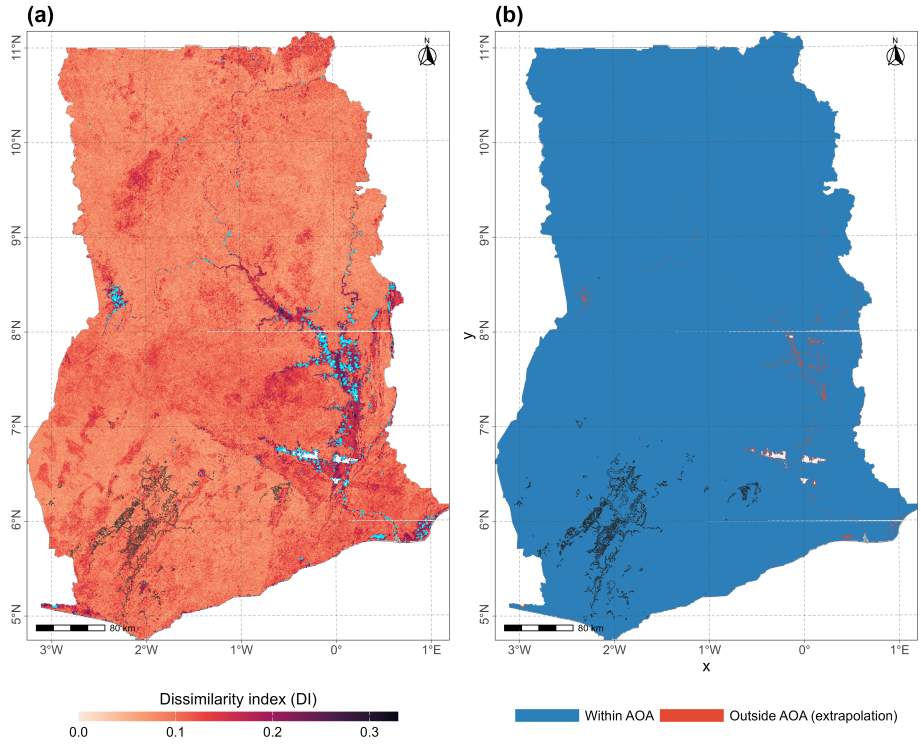


Figure 13. Area of Applicability map. Dissimilarity Index (DI) on the left (a) shows darker and cyan areas that fall outside of what the model has seen before. Area of applicability (AoA) map on the right (b). Treshold from CV cross-validated DI. Mining polygons outlined.

9. Composite data quality assessment

For assessment of composite data quality, I extracted valid pixel counts, residual cloud probability, temporal standard deviation of key bands (NIR, RED, SWIR1), and missing data coverage from 15,000 stratified random sample points across all 20 annual composites (See appendix Table 11, Figure 14). The composite for year 2000 shows the lowest data availability. Mean valid pixel counts of 2.3–3.7 across my early study period up to 2011 reflect the limited data availability and quality of the Landsat 5/7 archive for tropical West Africa at the time. The Coastal zone maintained marginal coverage (below 5 valid pixels) through 2011, improving substantially after Landsat 8 became operational in 2013. Residual cloud probability in the final composites was effectively zero across all zone-years, indicating that the medoid selection procedure successfully excluded cloud-contaminated observations. Extreme outliers were excluded through band clipping.

Table 11. Mean valid pixel count per composite by zone for selected years. Values below 5 (bold) indicate marginal coverage.

Year	Northern Savannah	Forest-Transition	Coastal
2000	3.6	3.4	**2.4**
2001	6.0	5.5	**4.3**
2005	6.4	5.7	**4.8**
2010	8.9	6.1	**3.7**
2013	14.9	13.5	9.3
2015	28.0	24.6	17.8
2019	33.8	28.4	22.0

Residual cloud probability was close to zero in all zone-years, but the low number of valid pixels in early years shows the limited Landsat data availability in cloudy areas.

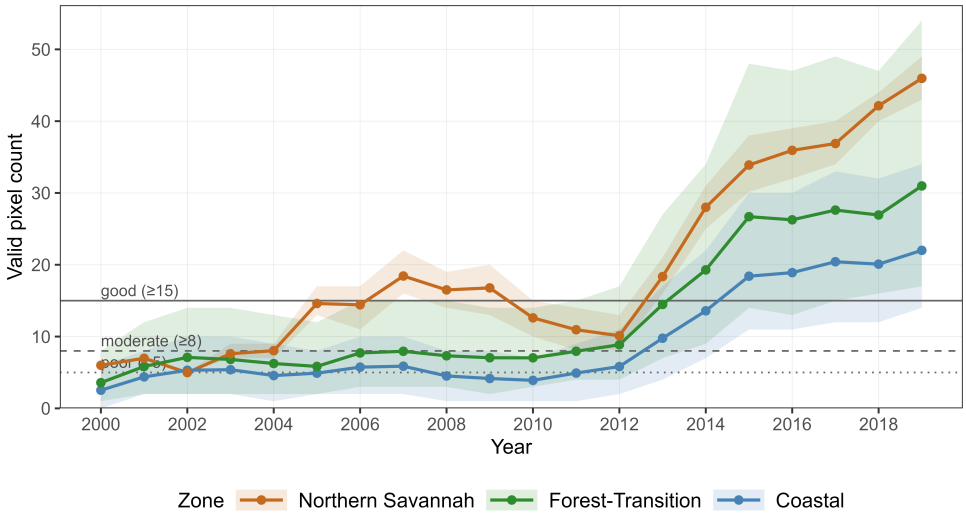


Figure 14. Valid pixel count per composite year within the study area. The step increase after 2012 reflects Landsat 8 availability. Horizontal lines indicate quality thresholds.

10. Industrial mining sites

To check whether the transition rule is too aggressive, I visualised four industrial mines with known start dates (Figure 15).

As visible, the RF and constraining approach manages to predict industrial mining onset reliably, even in earlier years.

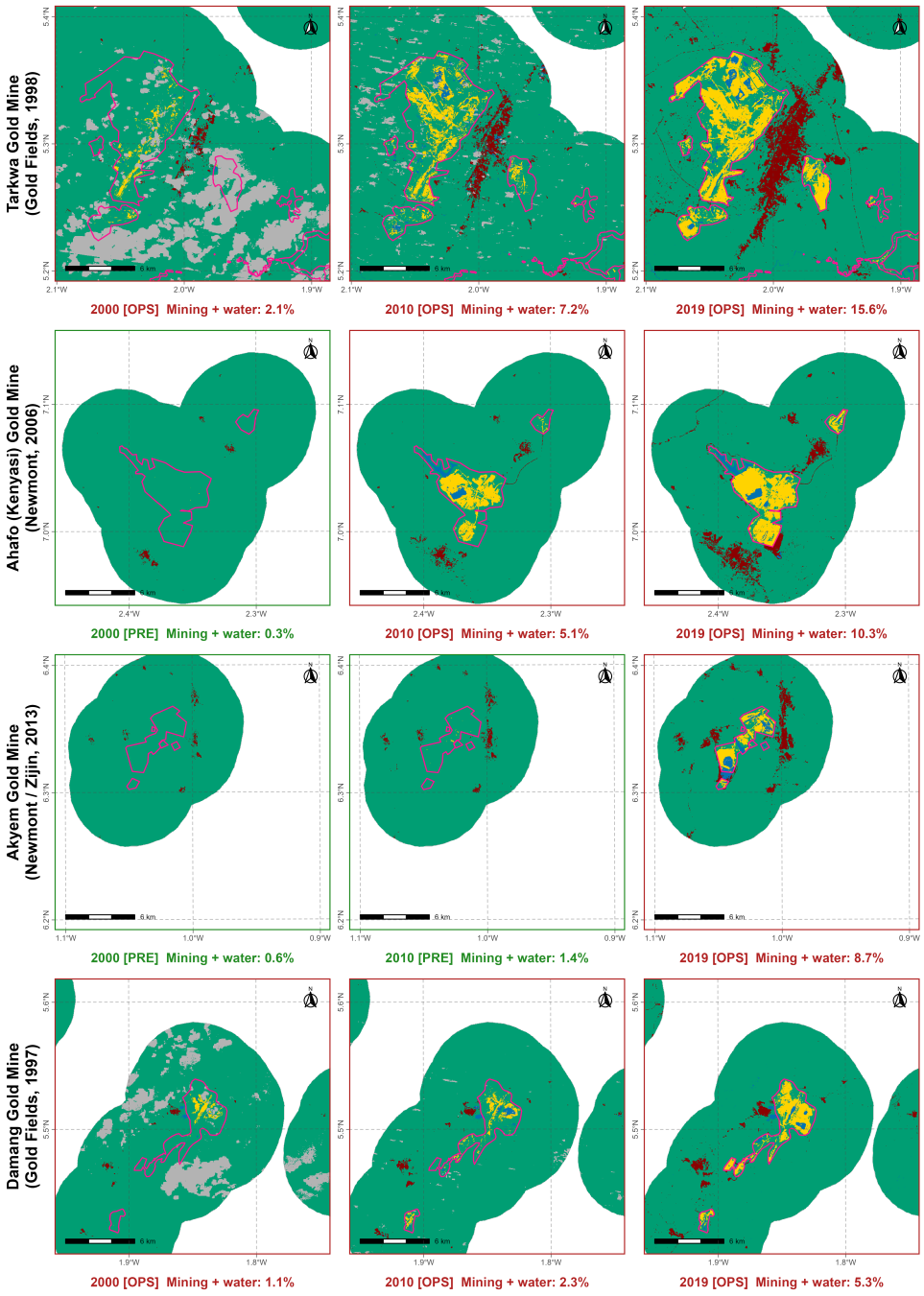


Figure 15. Constrained classification of four large industrial mines in 2000, 2010 and 2019, with mining polygons outlined. [PRE] = before, [OPS] = after the documented start of operations.

11. CNN validation of industrial mining onset

11.1. *CNN comparison*

As a consistency check, I trained a convolutional neural network (patch classifier)(LeCun, Bottou, Bengio, & Haffner, 1998) on 32x32 pixel tiles of the 2019 industrial

mine footprints and applied the model backwards to earlier years. I used footprints of 16 industrial mines from the Maus et al. (2022) polygons with a 5 km buffer around each mine and its Voronoi cell, so that mines did not share area. The CNN took six reflectance bands, eight spectral and teture indices, and a validaty mask for classification. The CNN consists of three convolution blocks (two 3x3 convolutions with batch normalisation and ReLu; (Ioffe & Szegedy, 2015)), used max-, and global average pooling (Lin, Chen, & Yan, 2014), dropout against overfitting (Srivastava, Hinton, Krizhevsky, Sutskever, & Salakhutdinov, 2014), and a sigmoid output. Whilst positive patches came from the industrial mining footprint of the 2019 composite, the negative examples came from mining cells outside the mining footprints (40%), surrounding landscape (20%), bright surfaces (20%), and random background (20%), so that the network had to also distinguish industrial from artisanal mining. Cross validation followed Roberts et al. (2017) with mines closer than 20km from each other grouped together and then assigned to four spatial folds, so the CNN could not memorise the mine it was dating. I calibrated year-specific detection thresholds for industrial mines on held-out imagery to assess sensor differences between Landsat 5/7 and 8 (See table 12). Mining onset (T10) was then defined as the first year in which at least 10% of a mine's 2019 mined extent (Mining and Water cells, vegetation excluded) was detected in three consecutive years. For the five mines established within the study period[12], RF onsets are compared to the CNN ones and the actual mine establishment and production information from the Ghana Minerals Commission (Ghana Minerals Commission, 2026a). The table below shows that the RF scored better than the CNN at mining onset detection.

Namdini mine which established in 2021 serves as a negative control here.

[12] E.g. Ahafo (2005).

Table 12. Onset of industrial mining per mine: random forest (RF) classification versus an independent convolutional neural network (CNN).

			2019 extent (ha)		Onset (T10)		
Mine	Est.	MinCom[a]	Mined	Threshold	RF	CNN	Δ[b]
Established 2001–2019							
Chirano	2005	≤2013	498	50	2005	2013	+8
Ahafo	2006	≤2013	1,971	197	2005	2006	+1
Edikan	2011	≤2013	1,148	115	2011	2012	+1
Nzema	2011	≤2013	260	26	2006	2011	+5
Akyem	2013	2013	752	75	2012	2004	−8
Established before 2000							
Nsuta	1916	≤2013	489	49	2004	2002	−2
Bogoso	1990	≤2013	427	43	2003	2004	+1
Iduapriem	1992	≤2013	871	87	≤2000	≤2000	–
Asanko	1997	2016	1,276	128	2012	≤2000	−12[c]
Bibiani	1997	≤2013	177	18	≤2000	2001	+1[c]
Damang	1997	≤2013	838	84	≤2000	≤2000	–
Tarkwa	1998	≤2013	2,482	248	2001	≤2000	−1[c]
Wassa	1998	≤2013	361	36	≤2000	2006	+6[c]
Excluded from CNN training and agreement statistics[d]							
Obuasi	1897	≤2013	42	10	≤2000	2002	+2[c]
Awaso	1941	≤2013	142	14	≤2000	n.d.	–
Prestea	1990	≤2013	398	40	2002	2006	+4

Notes: Est. = documented start of operations. Mined = 2019 RF mining and water cells inside the mine footprint (mining polygons within 5 km of the mine, split between neighbouring mines); vegetation inside the polygons is not counted. T10 = first year in which at least 10% of the 2019 mined extent (Threshold; minimum 10 ha) is detected in 3 consecutive observed years. CNN detection thresholds are calibrated per year to a 1% background false-positive rate. ≤ = left-censored (already met in the first observed year). n.d. = not detected.

[a] First reported large-scale production in Minerals Commission statistics (company-level data from 2013; ≤2013 = producing in the first reported month).

[b] CNN minus RF onset in years.

[c] One of the two onsets is left-censored, so the difference is a bound.

[d] Obuasi and Prestea are mainly underground operations; Awaso is the only bauxite mine. The negative control Namdini (established 2021) lies outside the RF prediction area.